HARI VIGNESH RAO 23BD1A052Q

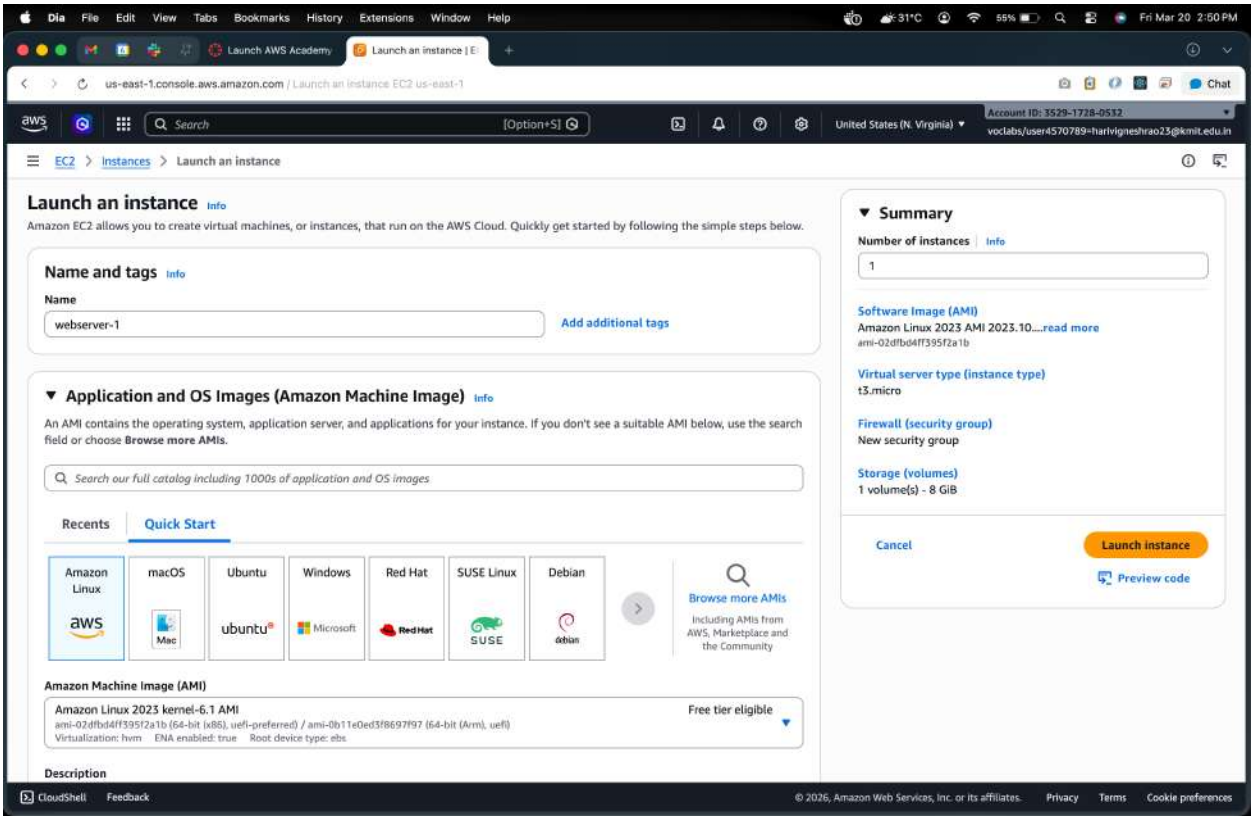
WEEK-9- Implementation of Elastic Load Balancer with Auto Scaling using AWS EC2 Instances.

Elastic Load Balancer (ALB)

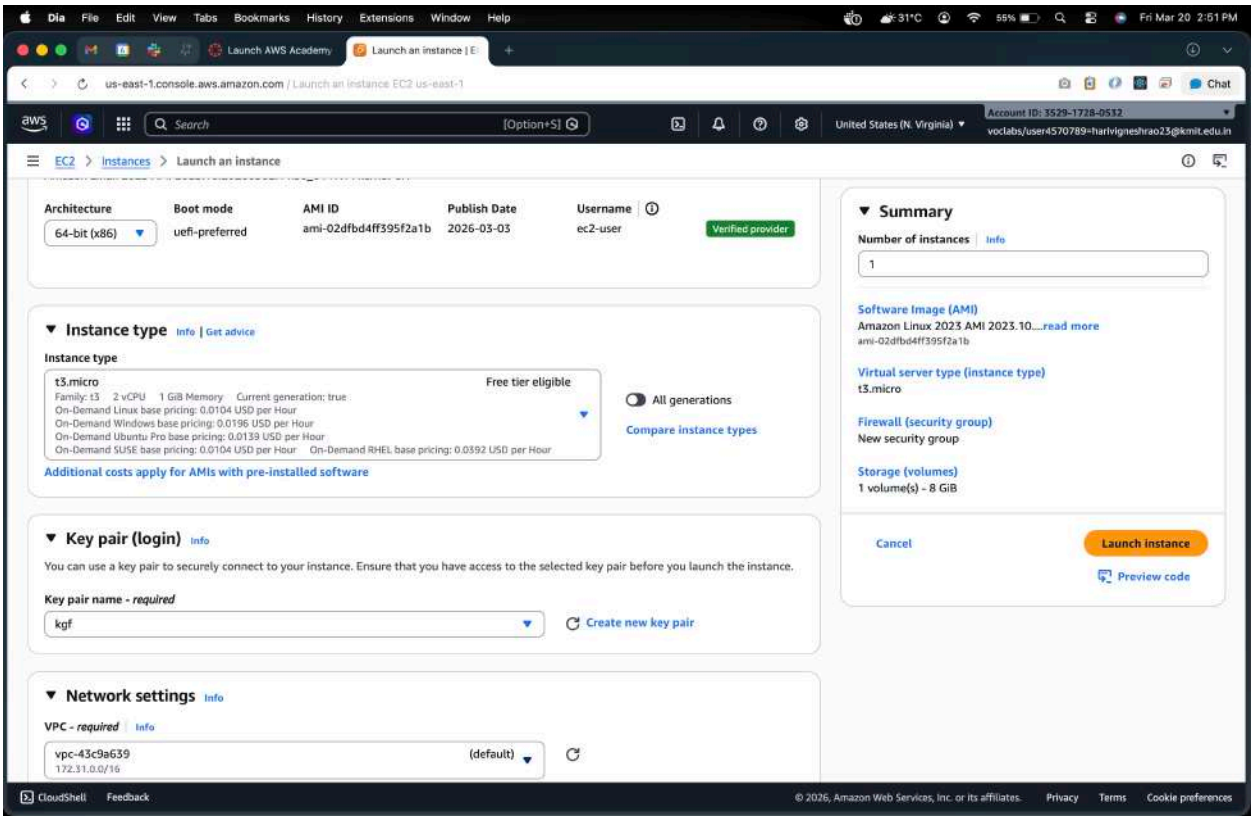
Step 1: Launch Two EC2 Instances

1.1 Launch EC2 Instance #1

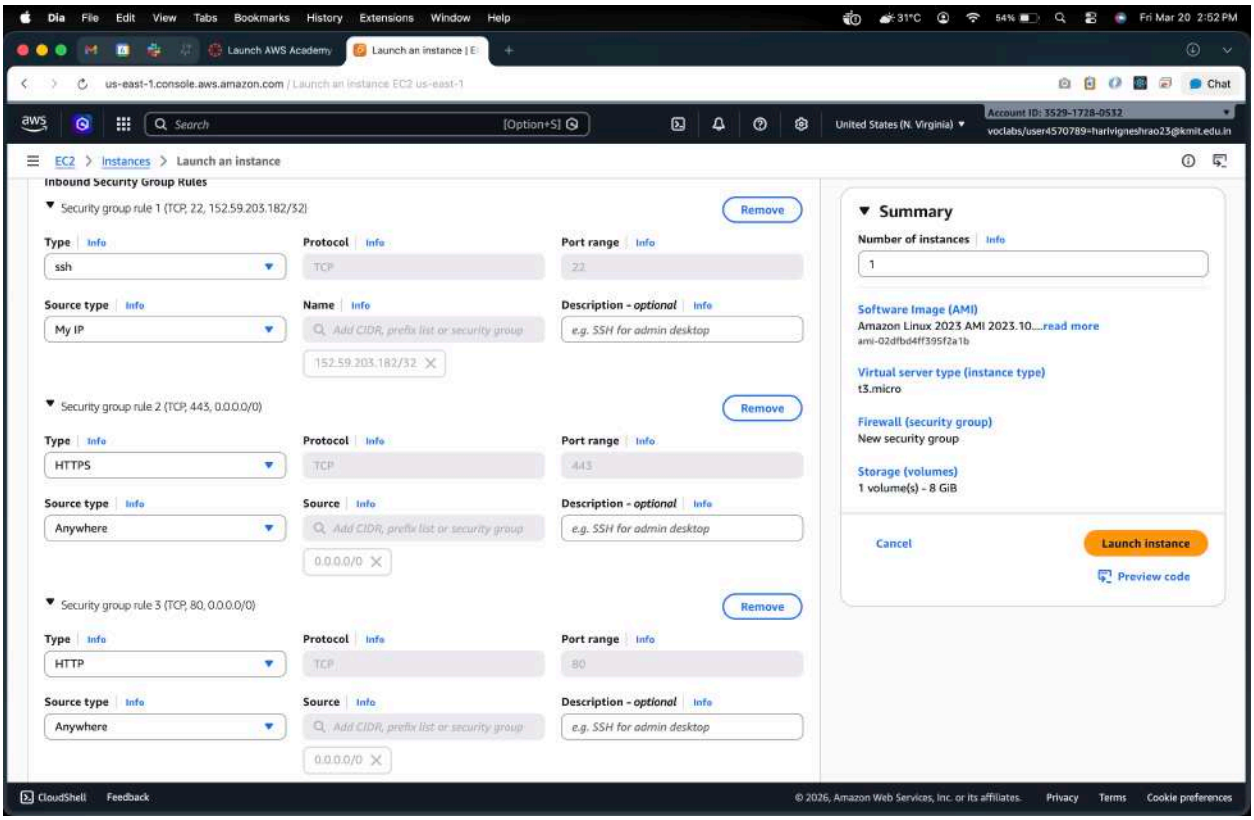
1. Go to **AWS EC2 Console → Launch Instance**.
2. **Name**: webserver-1
3. Choose an **AMI** (Amazon Linux 2).



4. Select **Instance Type** (e.g., t2.micro).
5. Add **Storage** (default is fine).



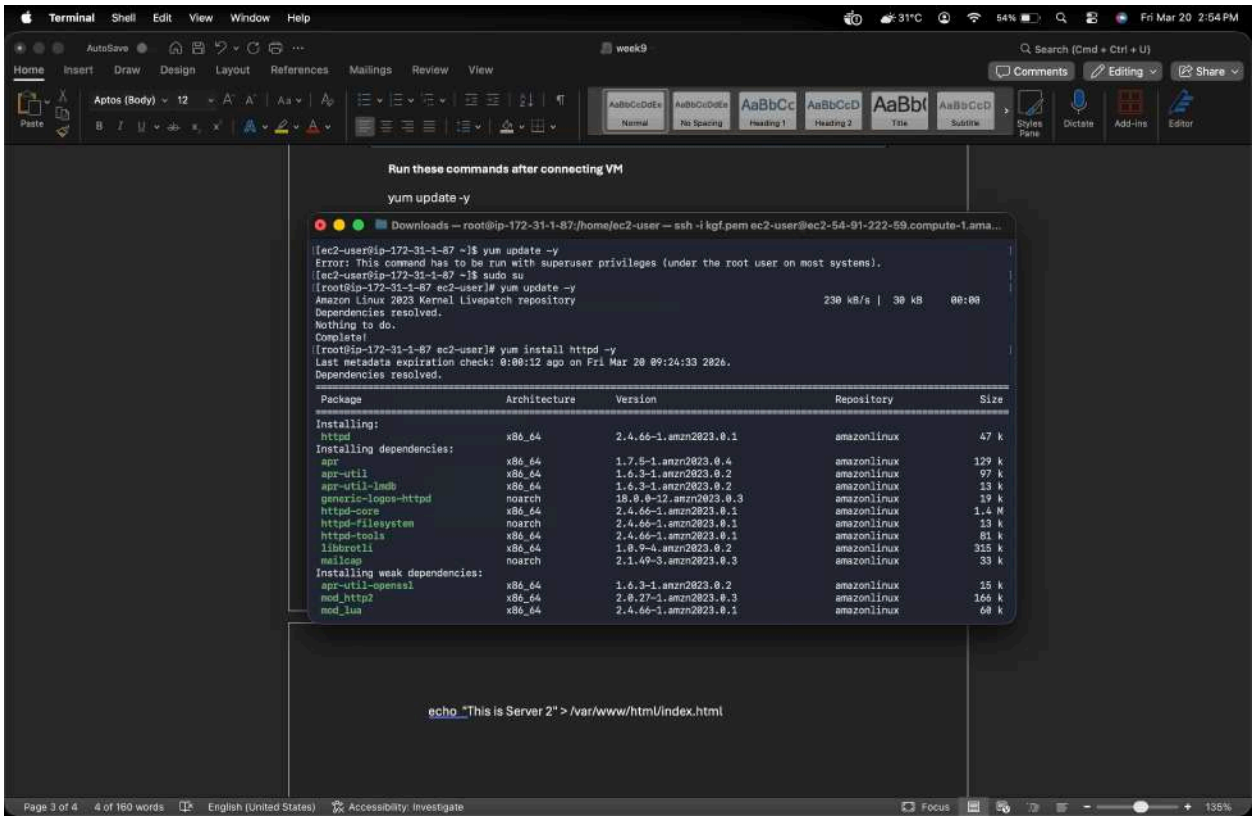
6. Configure **Security Group**:
    - Allow **SSH (22)** from your IP
    - Allow **HTTP (80)** from anywhere (0.0.0.0/0)
7. **Launch** → Select/Create Key Pair → Launch.



Run these commands after connecting VM

yum update -y

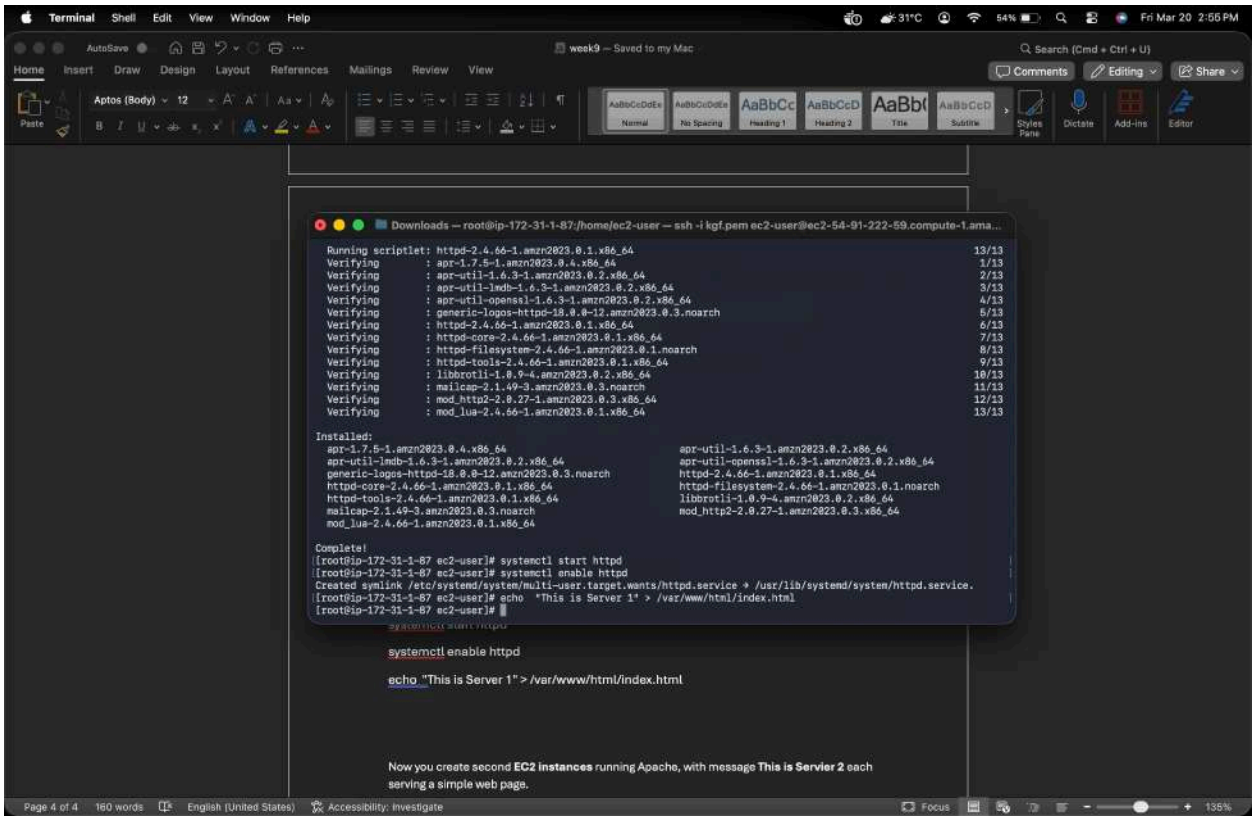
yum install httpd -y



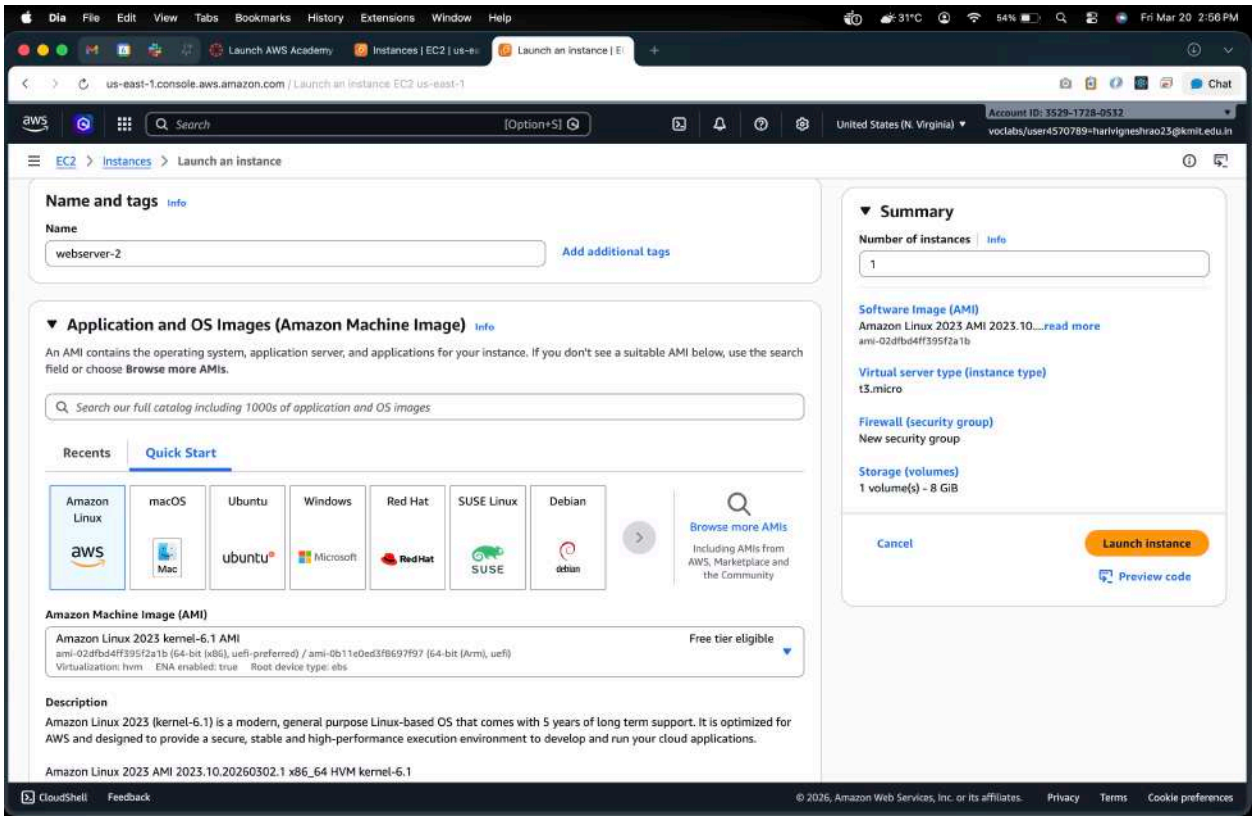
systemctl start httpd

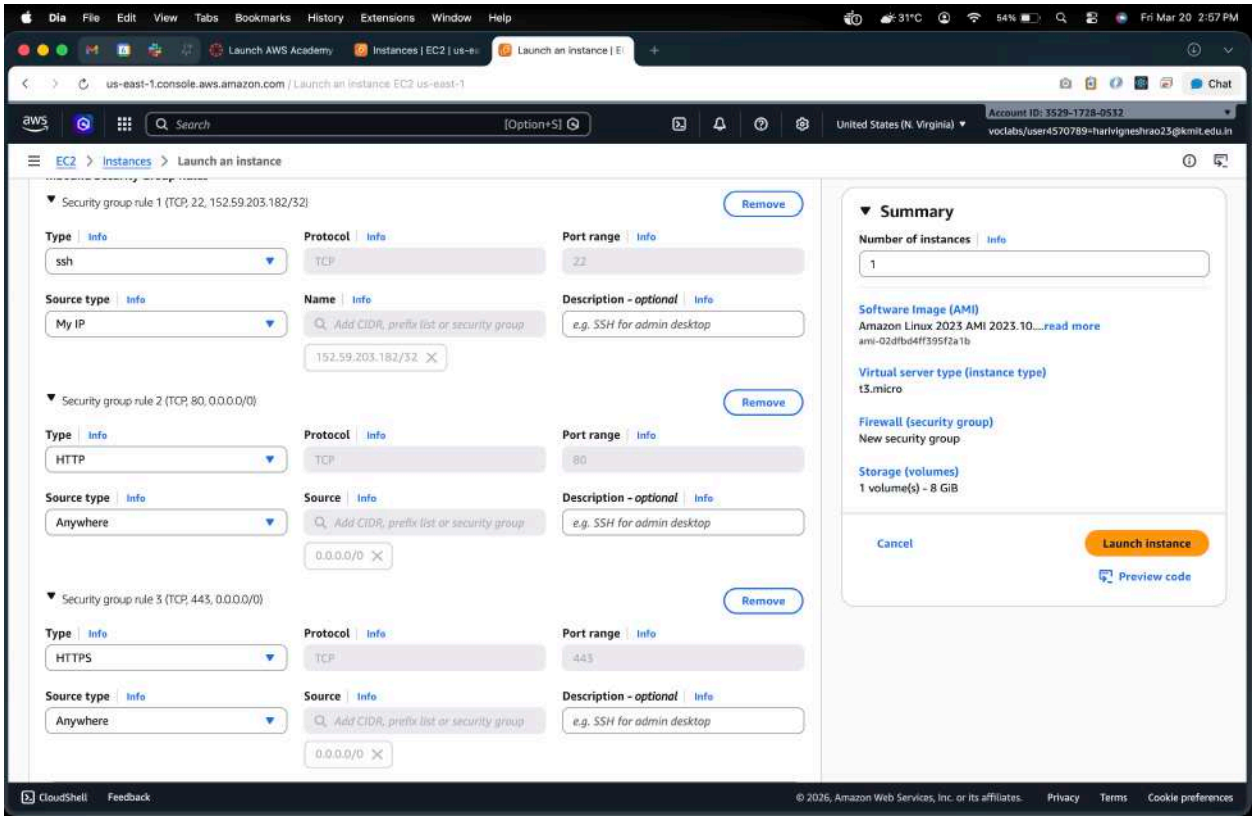
systemctl enable httpd

echo "This is Server 1" > /var/www/html/index.html



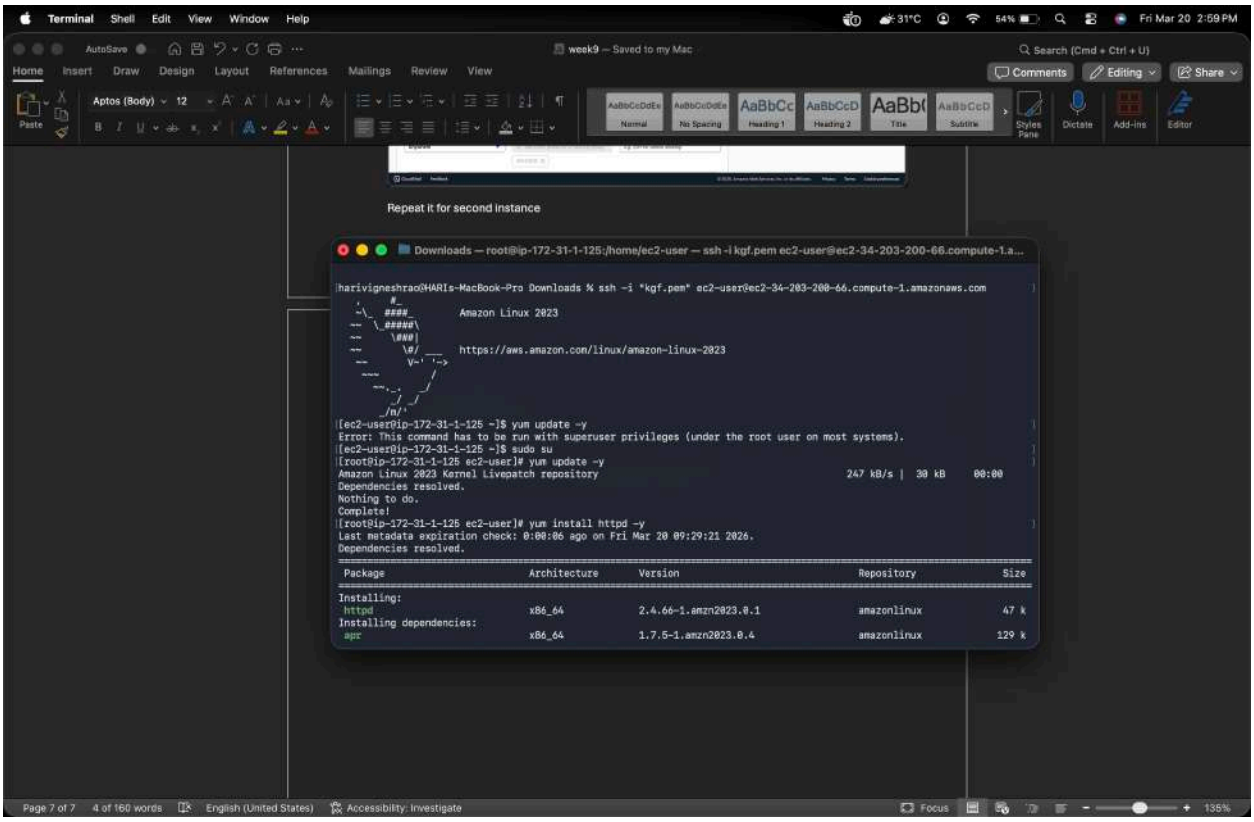
Now you create second **EC2 instances** running Apache, with message **This is Servier 2** each serving a simple web page.





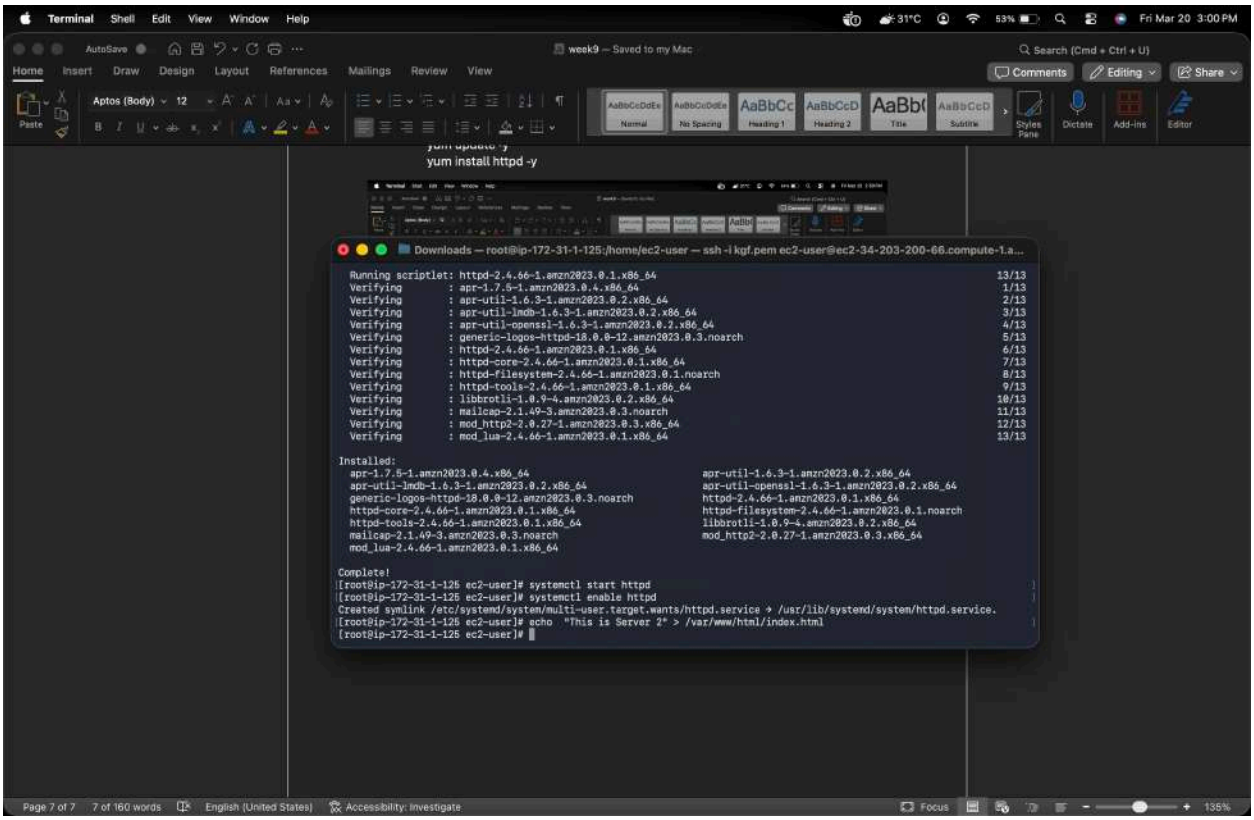
Repeat it for second instance

yum update -y
yum install httpd -y



systemctl start httpd
systemctl enable httpd

echo "This is Server 2" > /var/www/html/index.html

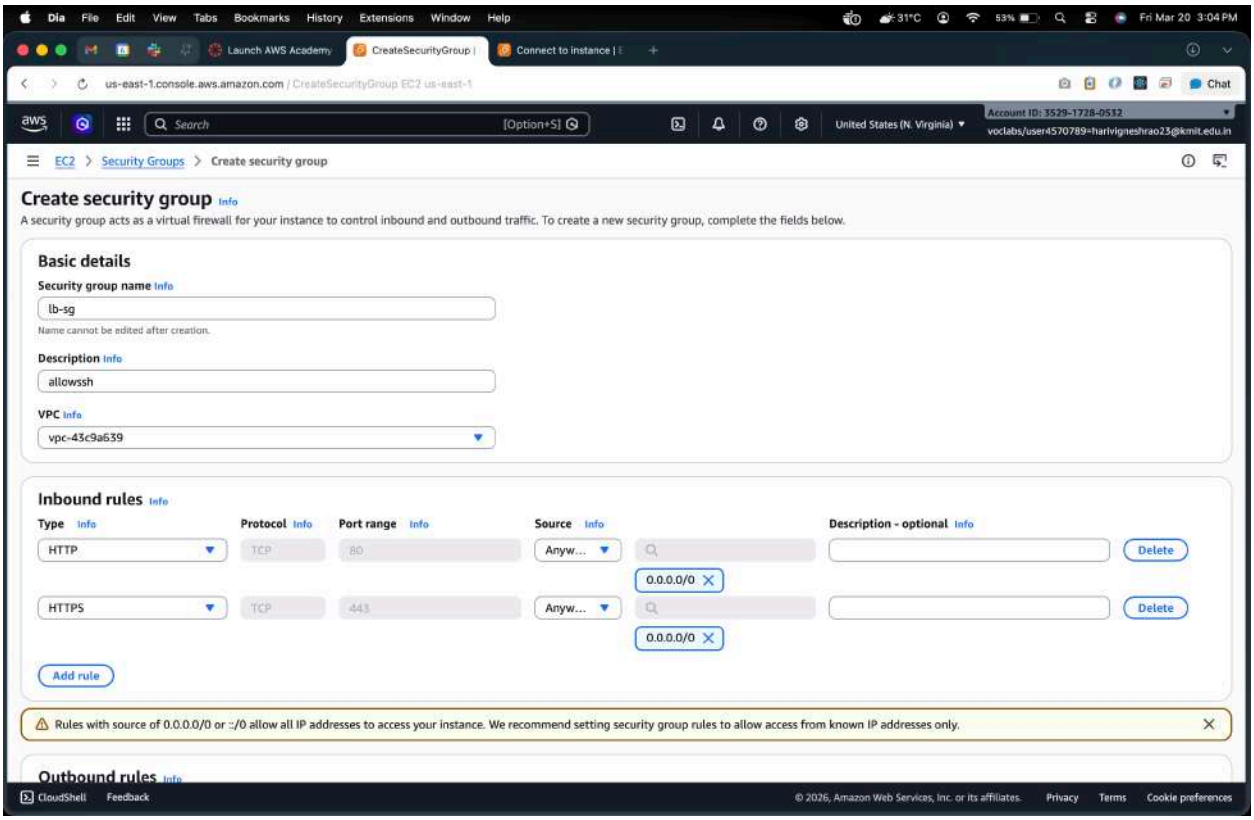


Step 2: Create a Load Balancer

AWS provides **Elastic Load Balancer (ELB)**. We will create an **Application Load Balancer (ALB)**.

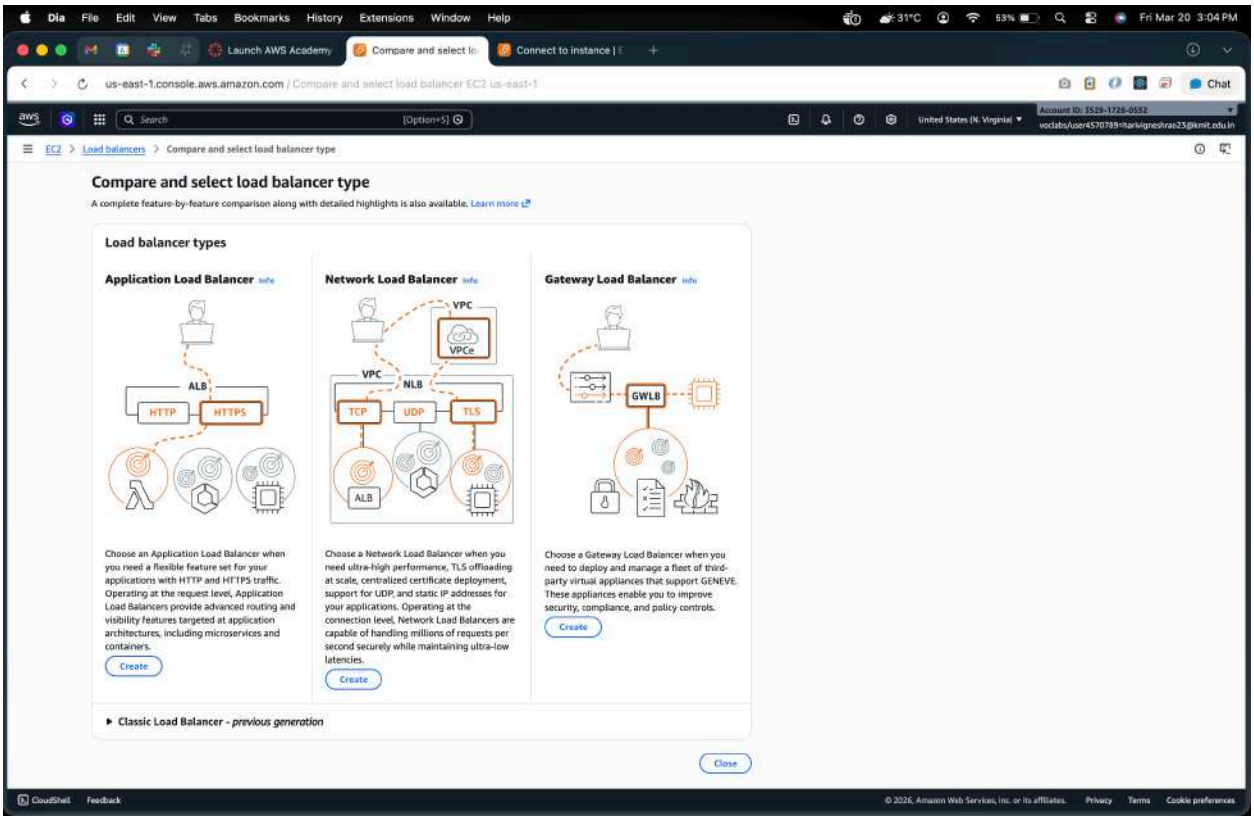
2.1 Configure Security Group for Load Balancer

- Go to **EC2 → Security Groups → Create Security Group**
    - Name: lb-sg
    - Allow **HTTP (80)** from anywhere
    - Optional: HTTPS (443) if using SSL

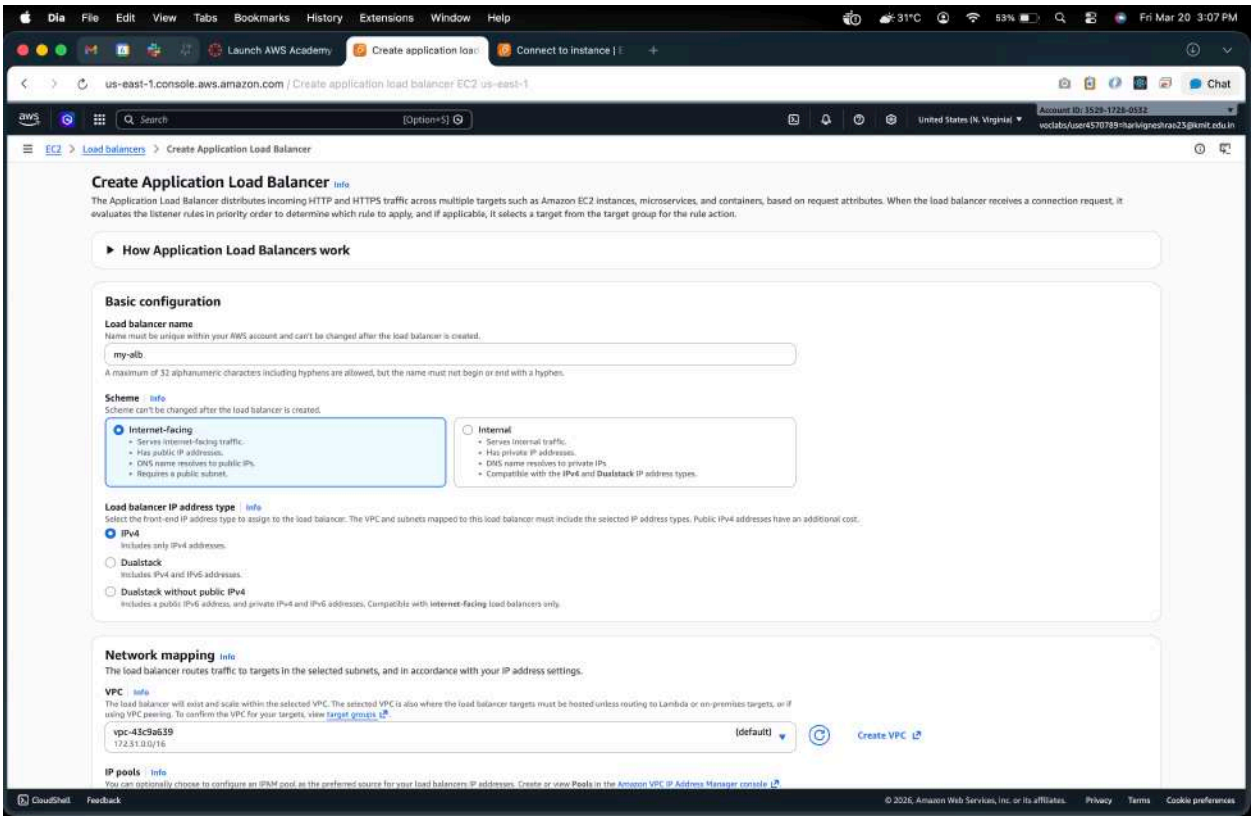


2.2 Create the Load Balancer

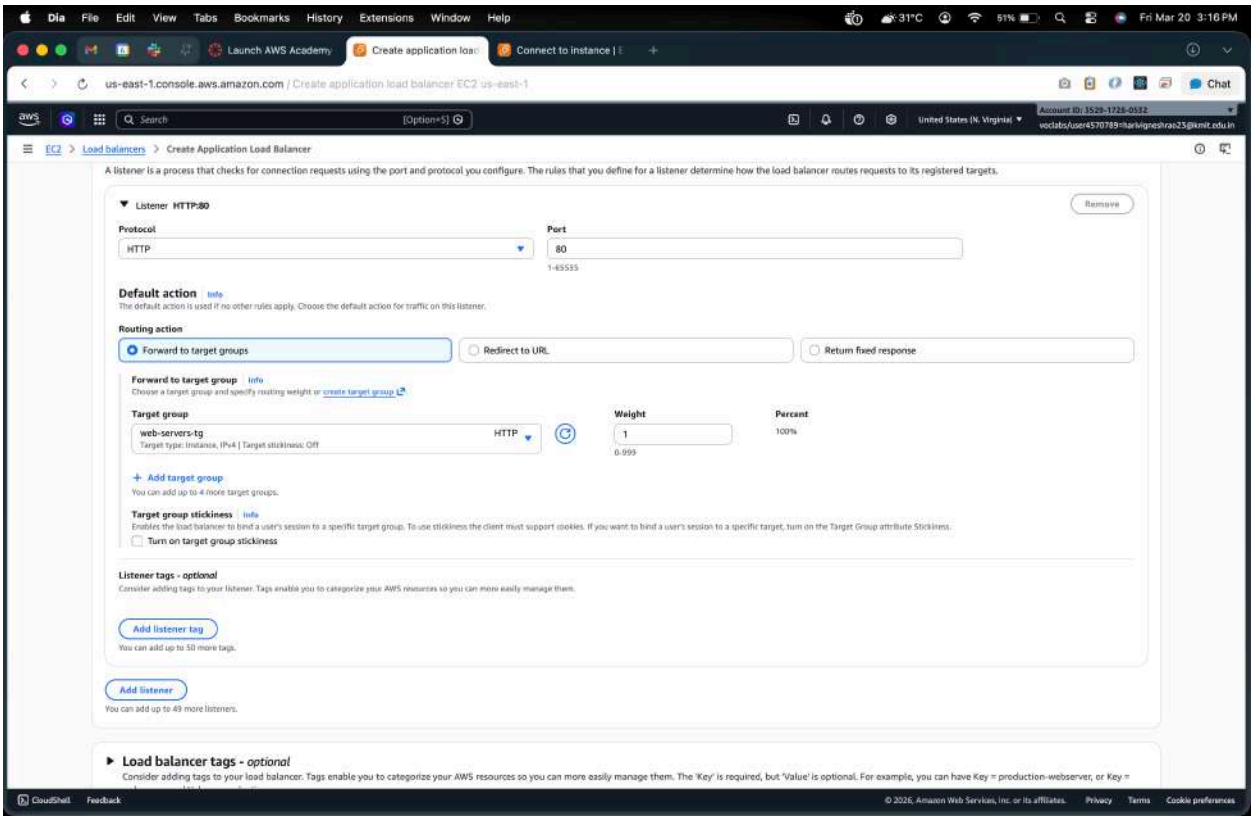
1. Go to **EC2 → Load Balancers → Create Load Balancer → Application Load Balancer**



2. **Basic Configuration**:
    - Name: my-alb
    - Scheme: Internet-facing
    - IP address type: IPv4

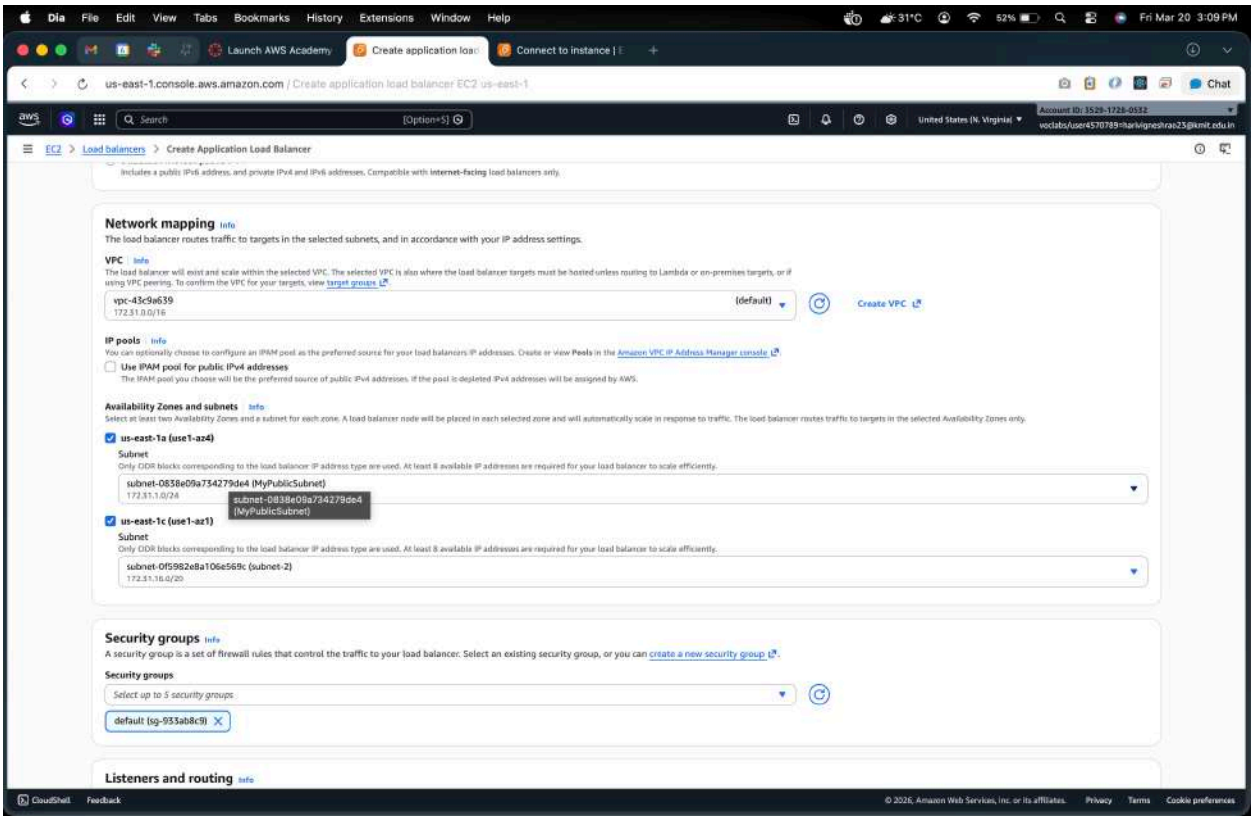


3. **Listeners**:
    - Default: HTTP (80)



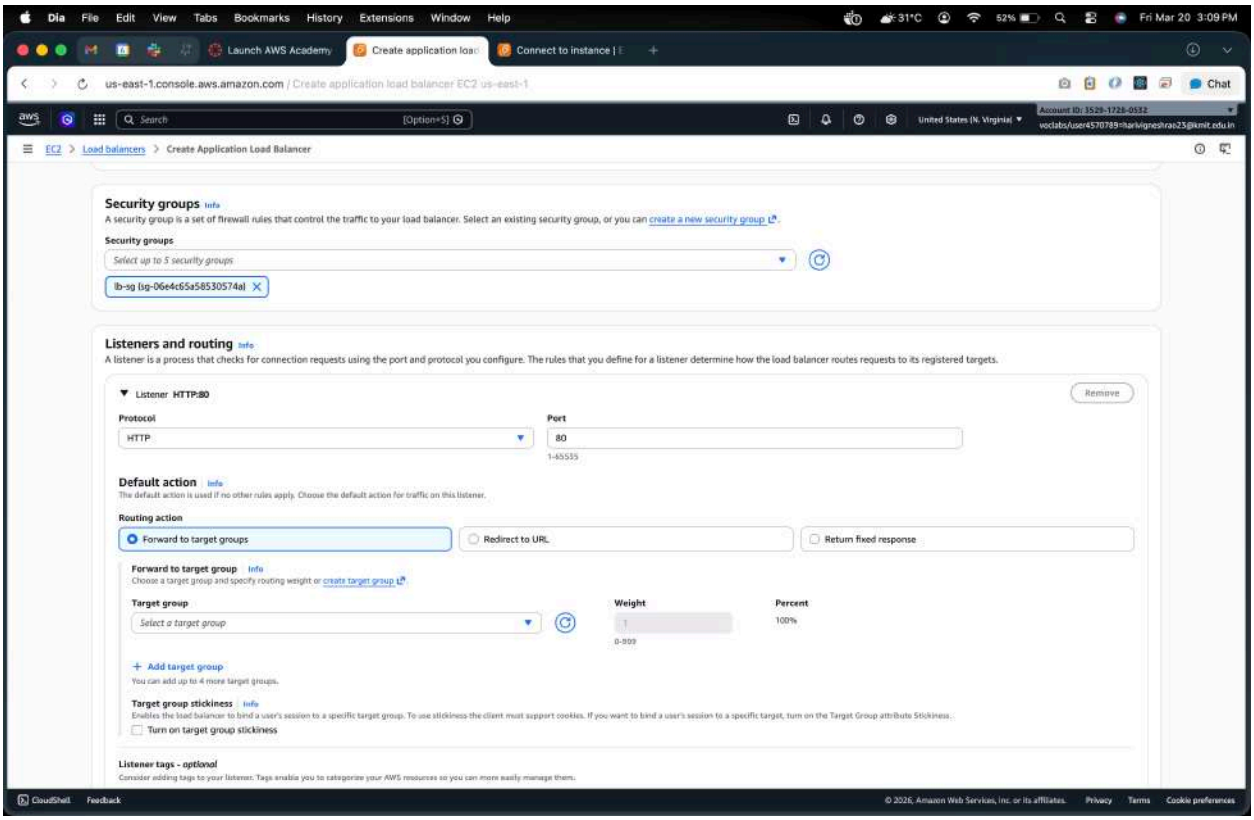
2.3 Configure Availability Zones

- Select the deafault **VPC** where your EC2 instances exist
- Select **two /all subnets** (one for each instance if possible)



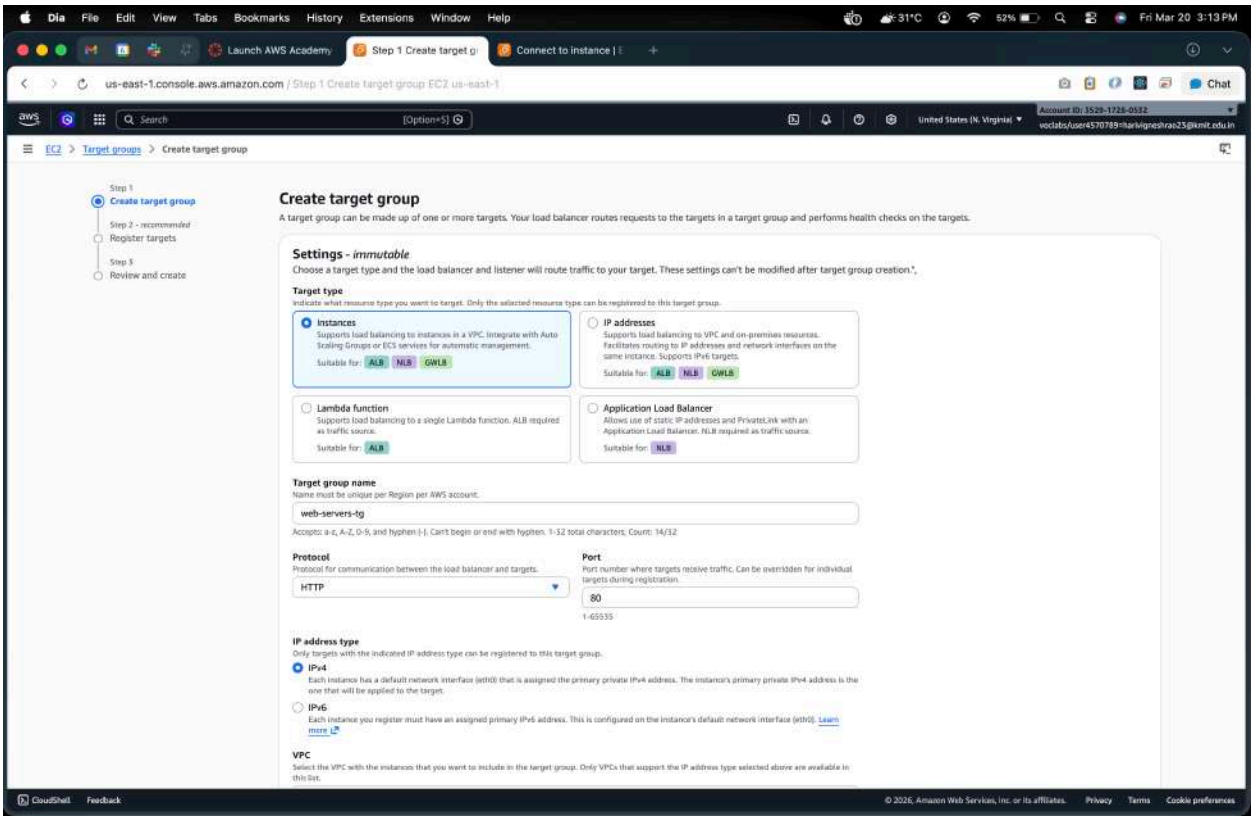
2.4 Configure Security Groups

- Choose the **lb-sg** security group created earlier

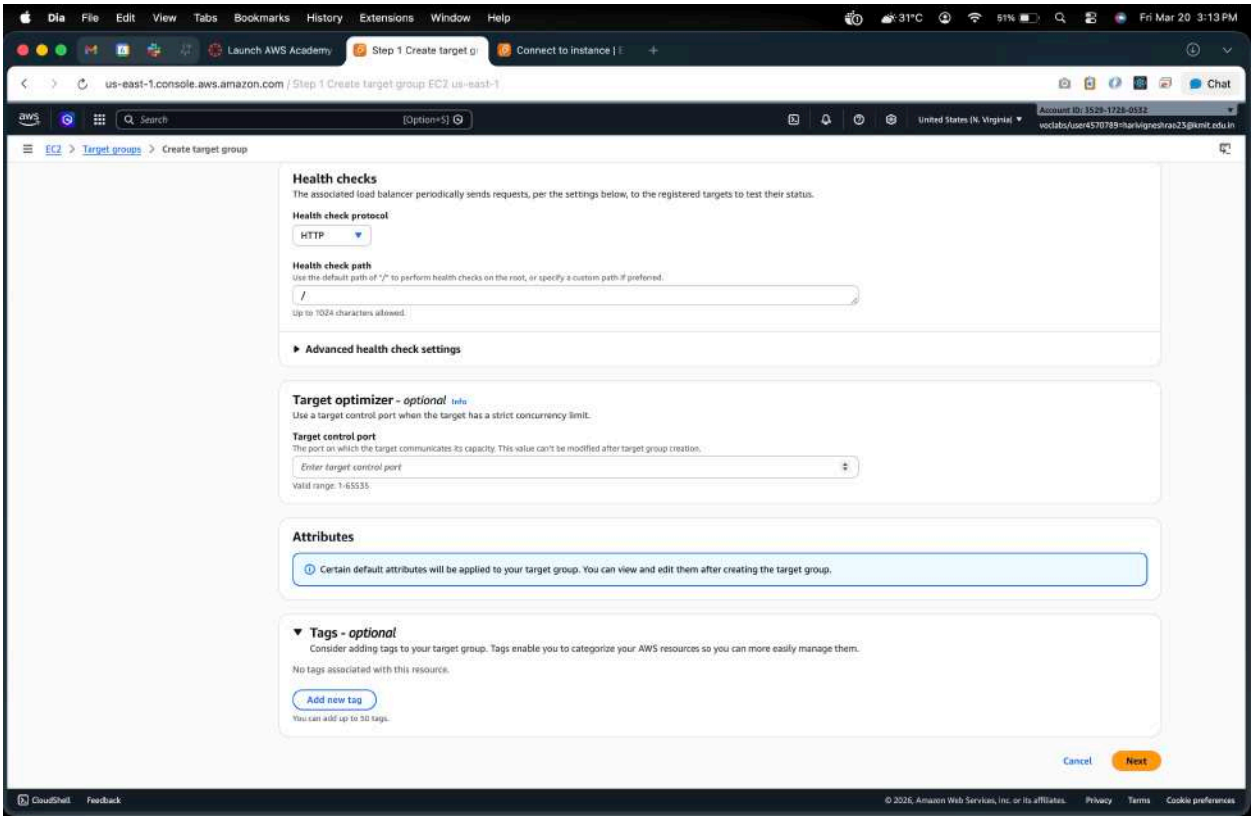


2.5 Configure Target Groups

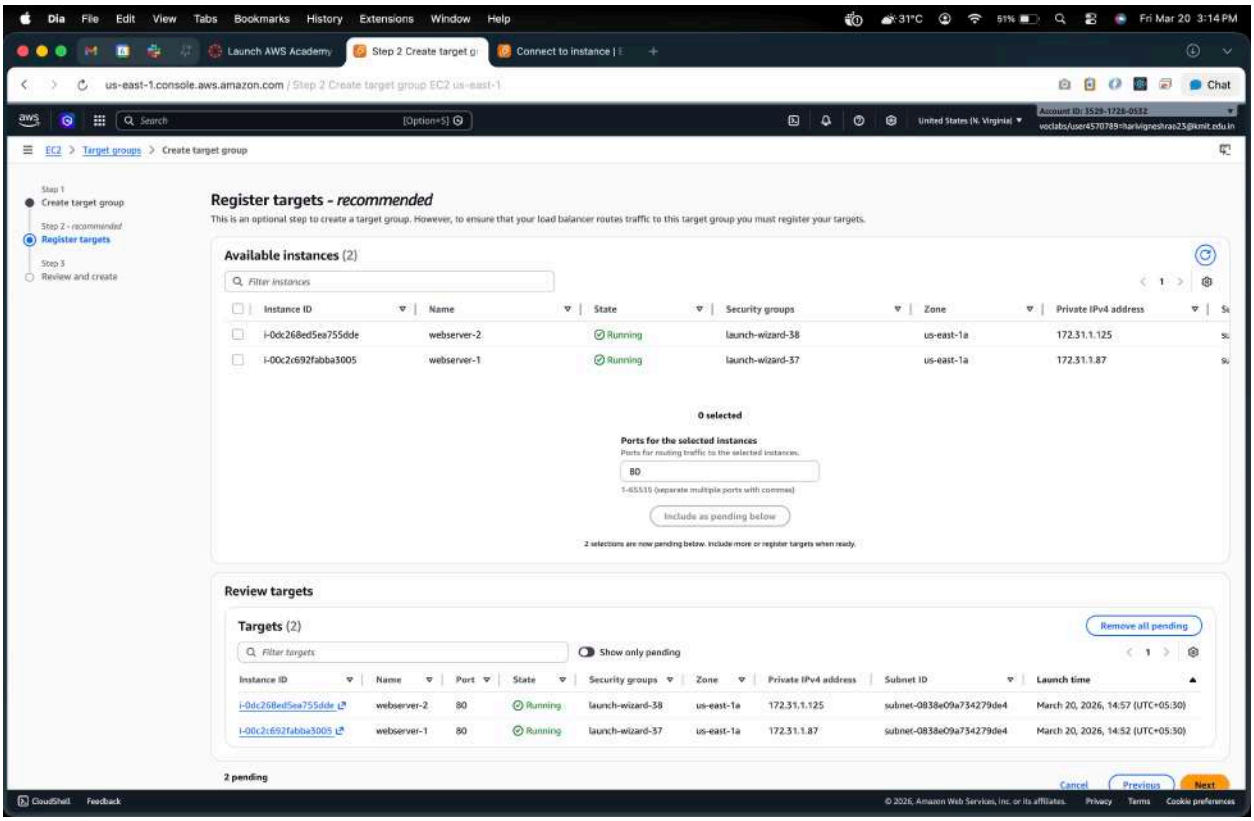
1. **Create a new target group**:
    - Name: web-servers-tg
    - Target type: Instance
    - Protocol: HTTP
    - Port: 80

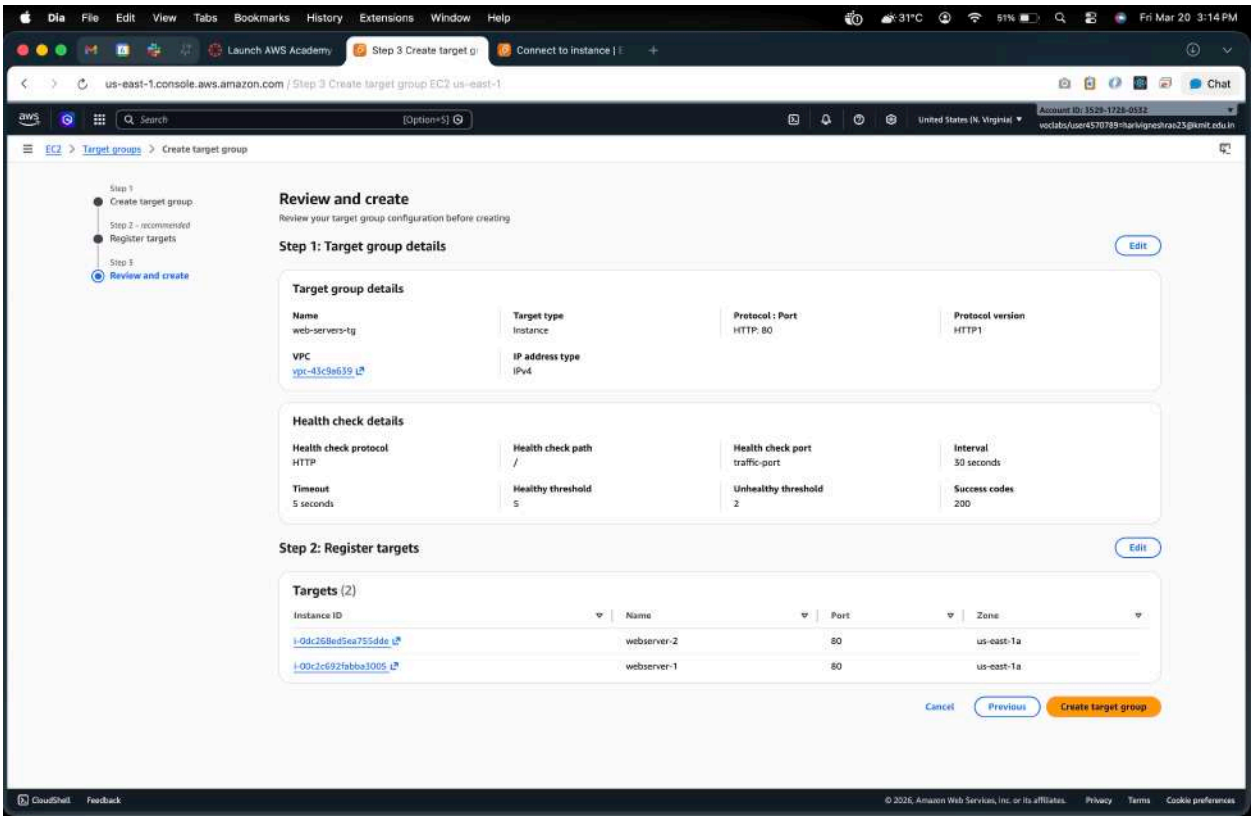


- Health checks: / (default)



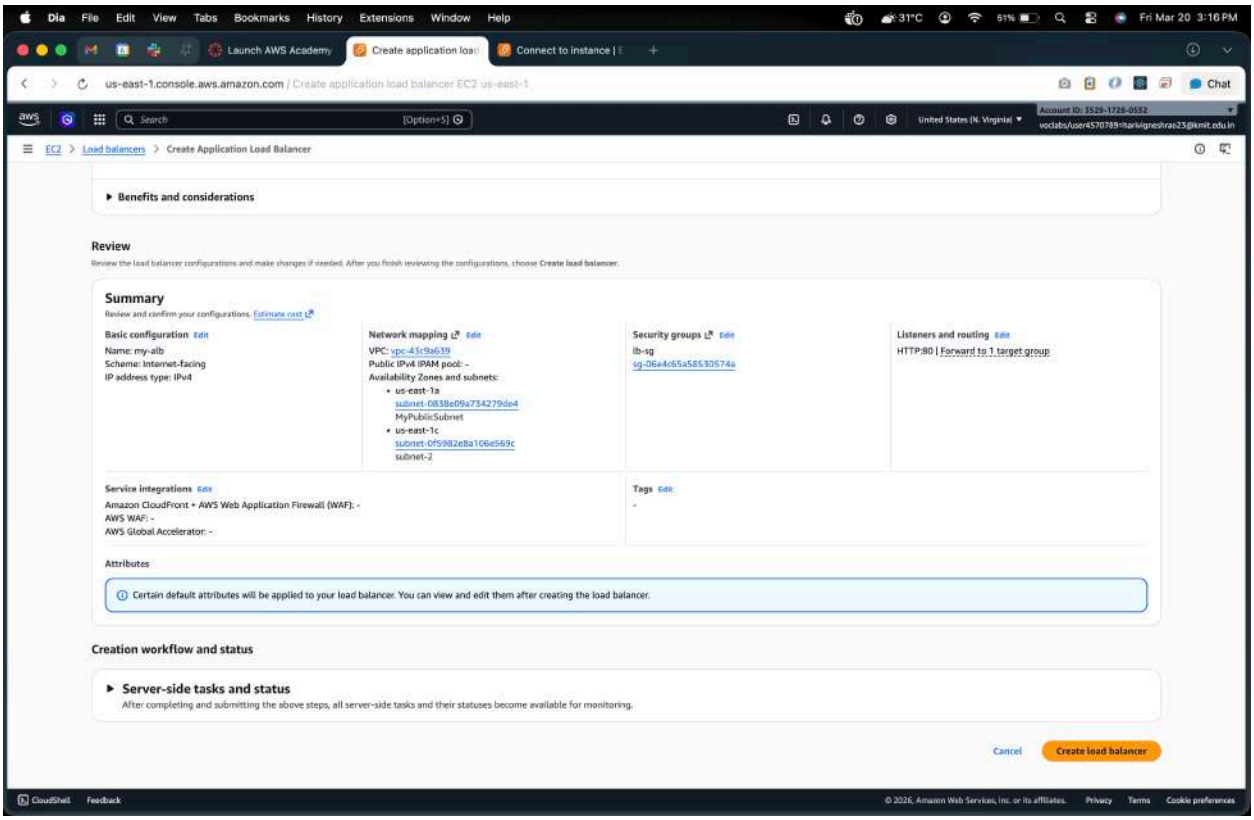
2. **Register Targets**:
    - Select EC2 INSTANCES HERE --- **webserver-1** and **webserver-2**
    - Click **Include as pending → Register**





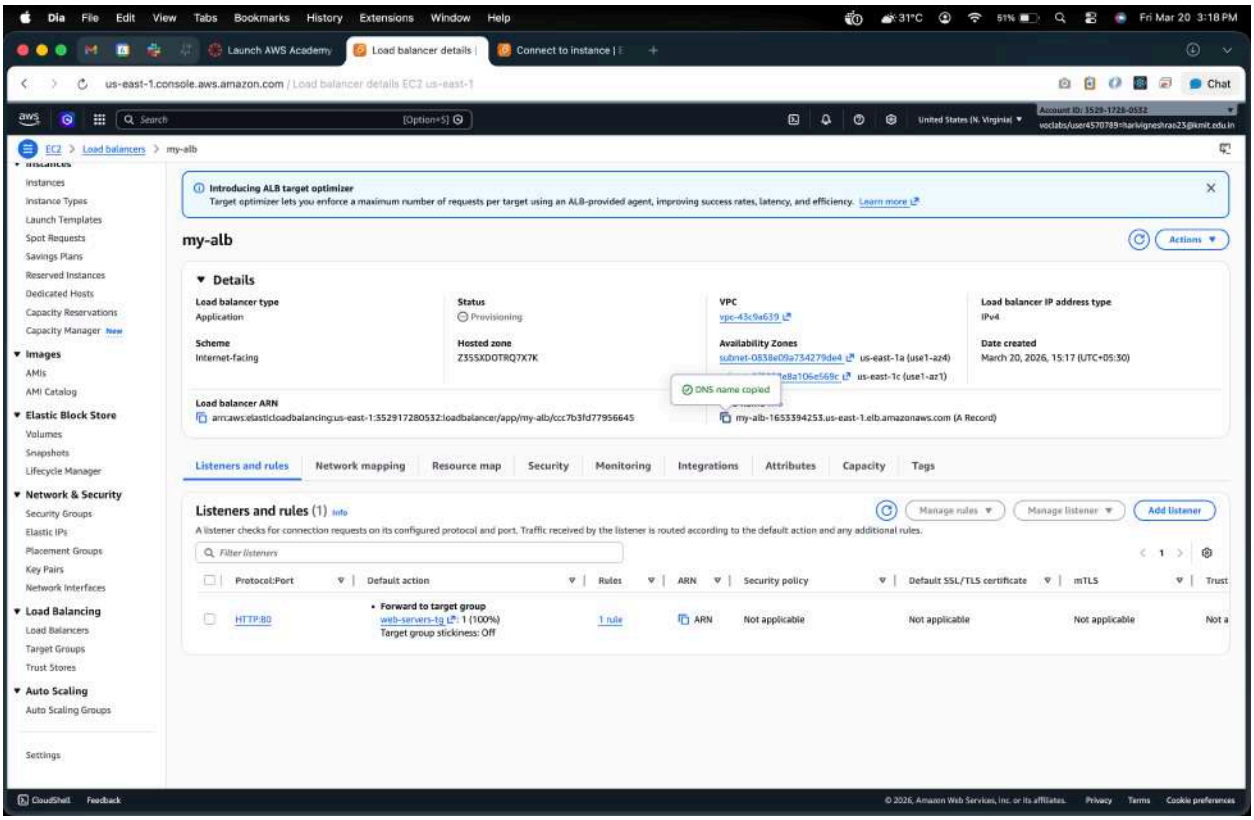
2.6 Review and Create

- Review all settings → Click **Create Load Balancer**
- AWS will take a few minutes to provision the ALB

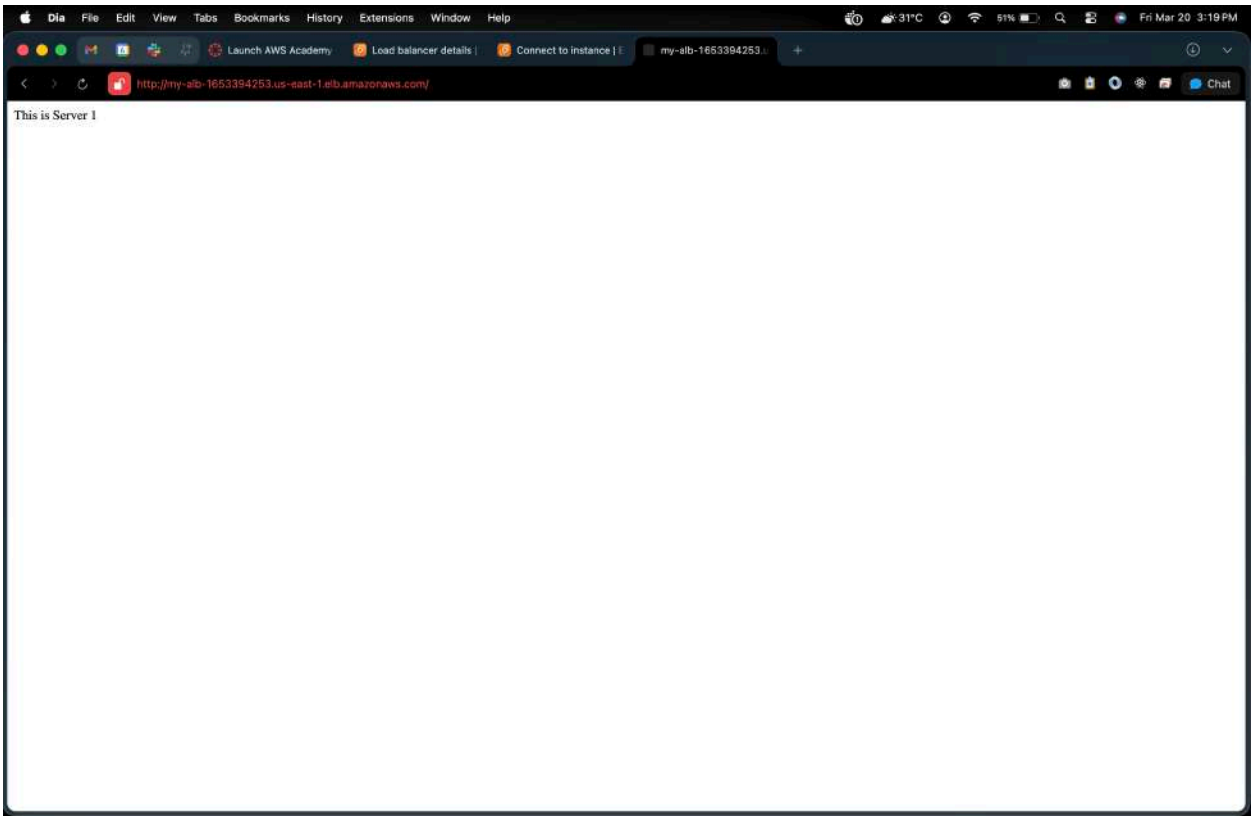


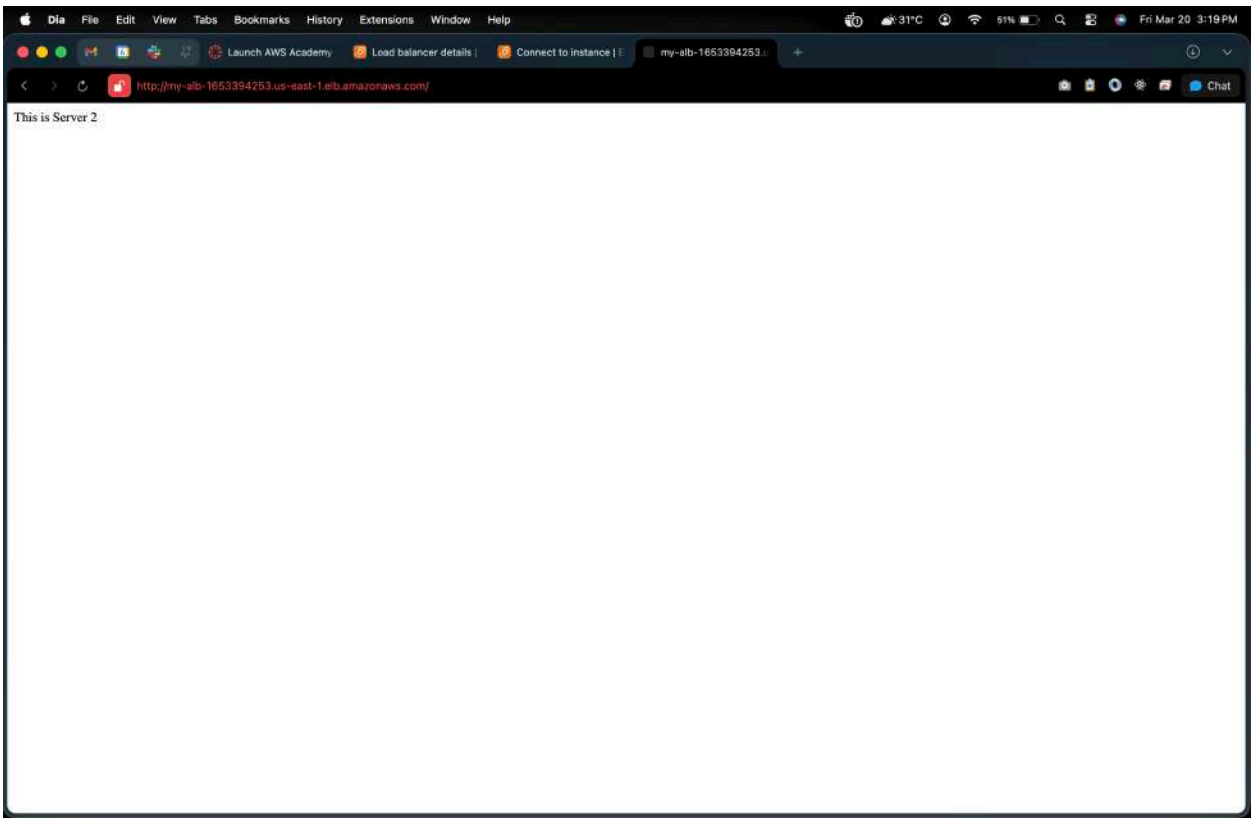
Step 3: Verify Load Balancer

1. Go to **Load Balancers → Select ALB → Description → DNS name**
2. Copy the **DNS name** (something like my-alb-123456.us-east-1.elb.amazonaws.com)



3. Paste it in a browser → You should see either **WebServer-1** or **WebServer-2** page.
    - Refresh multiple times → Load balancer distributes traffic between the two instances

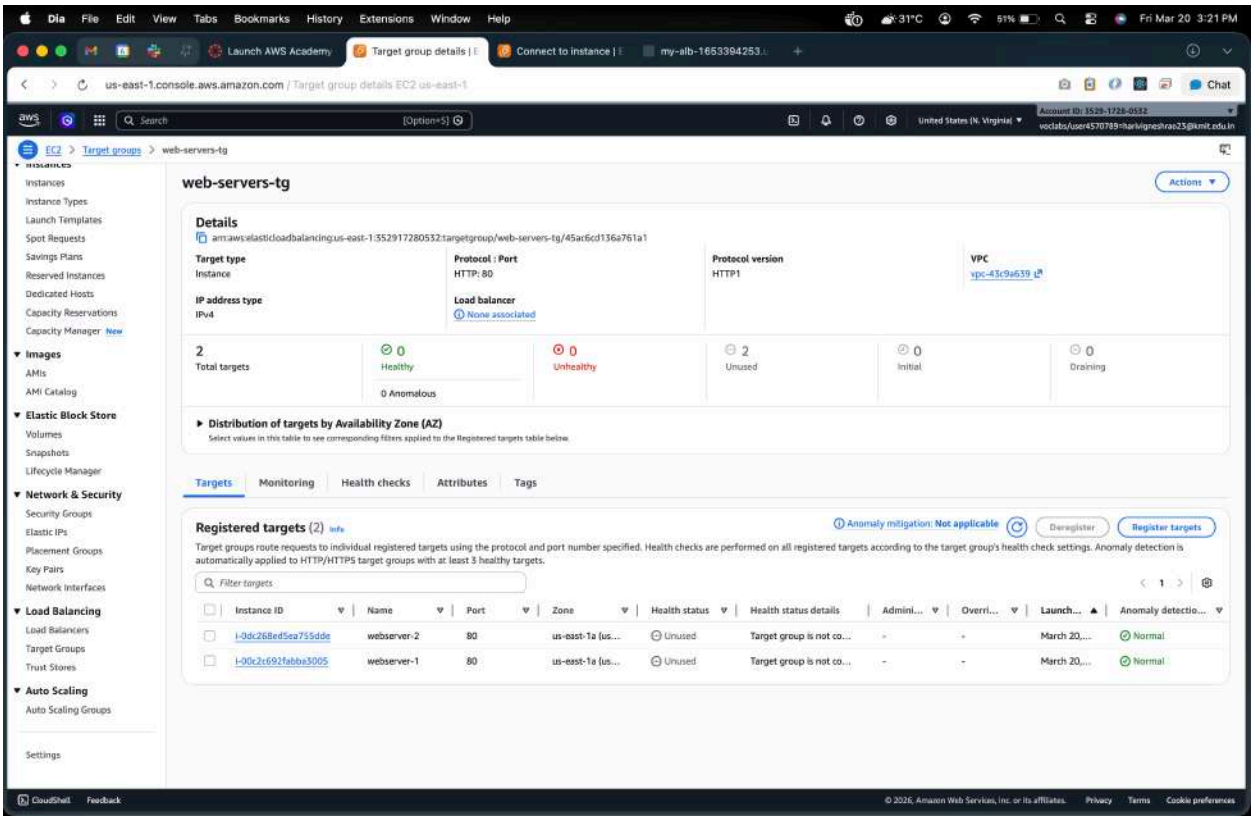

Dia File Edit View Tabs Bookmarks History Extensions Window Help
31°C
51%
Fri Mar 20 3:19 PM
Launch AWS Academy
Load balancer details |
Connect to instance | I
my-alb-1653394253.
http://my-alb-1653394253.us-east-1.elb.amazonaws.com/
Chat
This is Server 1


Dia File Edit View Tabs Bookmarks History Extensions Window Help
31°C
51%
Fri Mar 20 3:19 PM
Launch AWS Academy
Load balancer details |
Connect to instance | I
my-alb-1653394253.
http://my-alb-1653394253.us-east-1.elb.amazonaws.com/
Chat
This is Server 2

Step 4: Optional – Test Health Checks

- Go to **Target Groups → web-servers-tg → Targets**
- Ensure **Status** of both instances is healthy

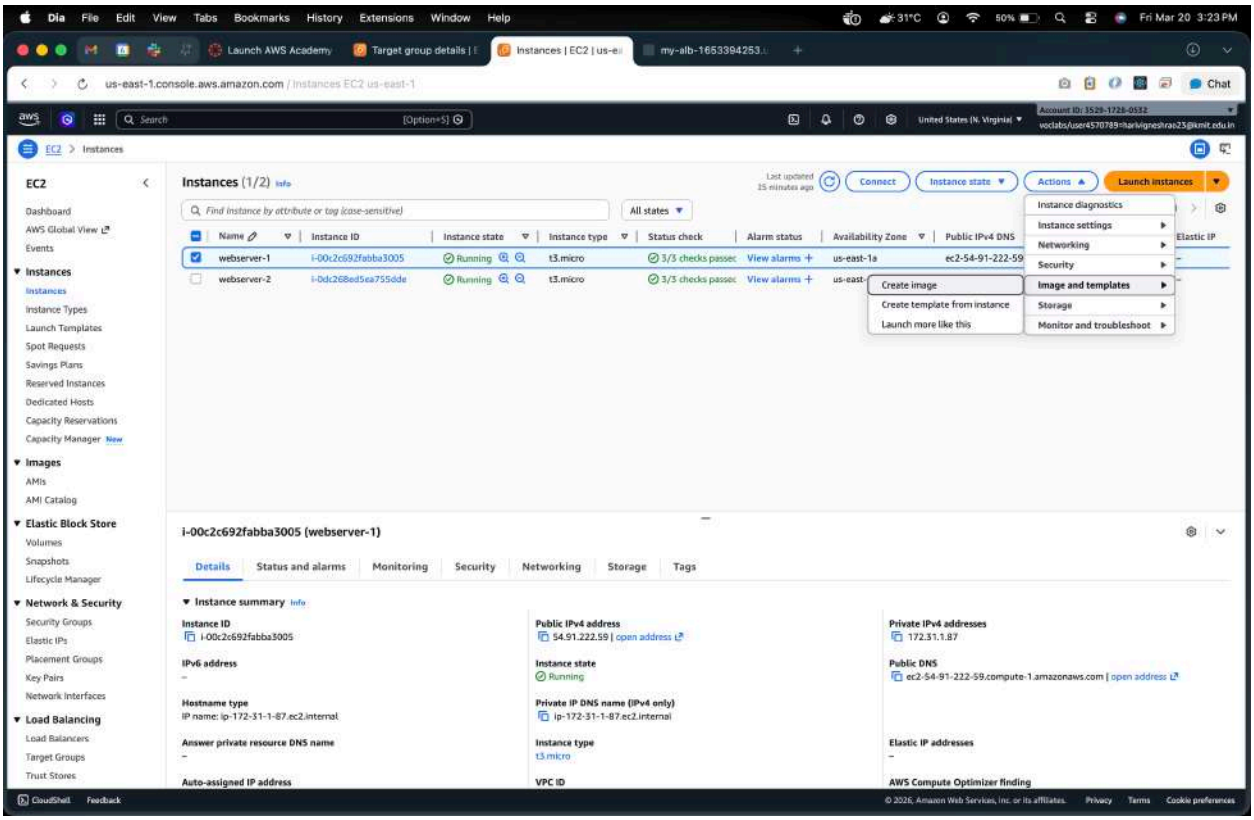
If a server is unhealthy, check the bootstrapping script or security group rules.



Step-by-Step: Auto Scaling Group with Load Balancer

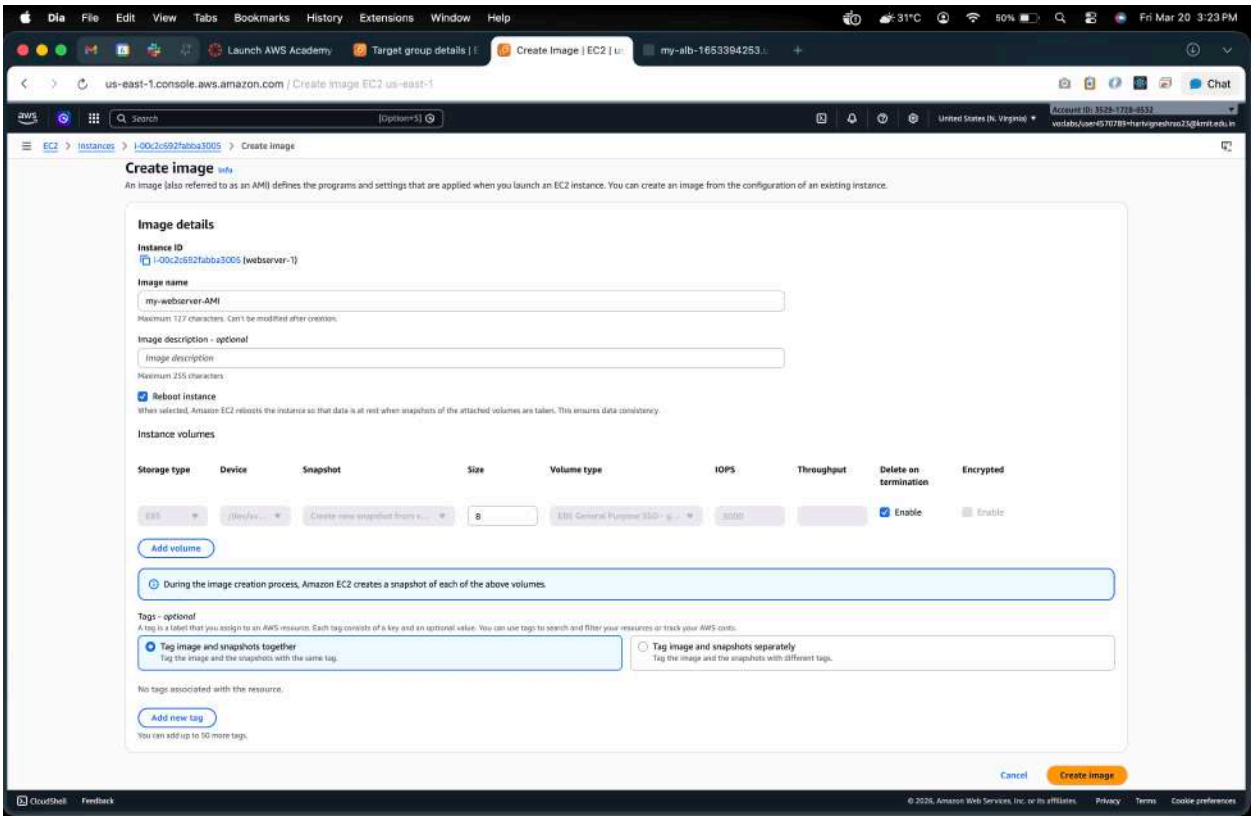
Step 1: Create an AMI from Existing EC2 Instance

1. Go to **EC2 Console → Instances**
2. Select the EC2 instance you already configured with application
3. Click **Actions → Image and Templates → Create Image**



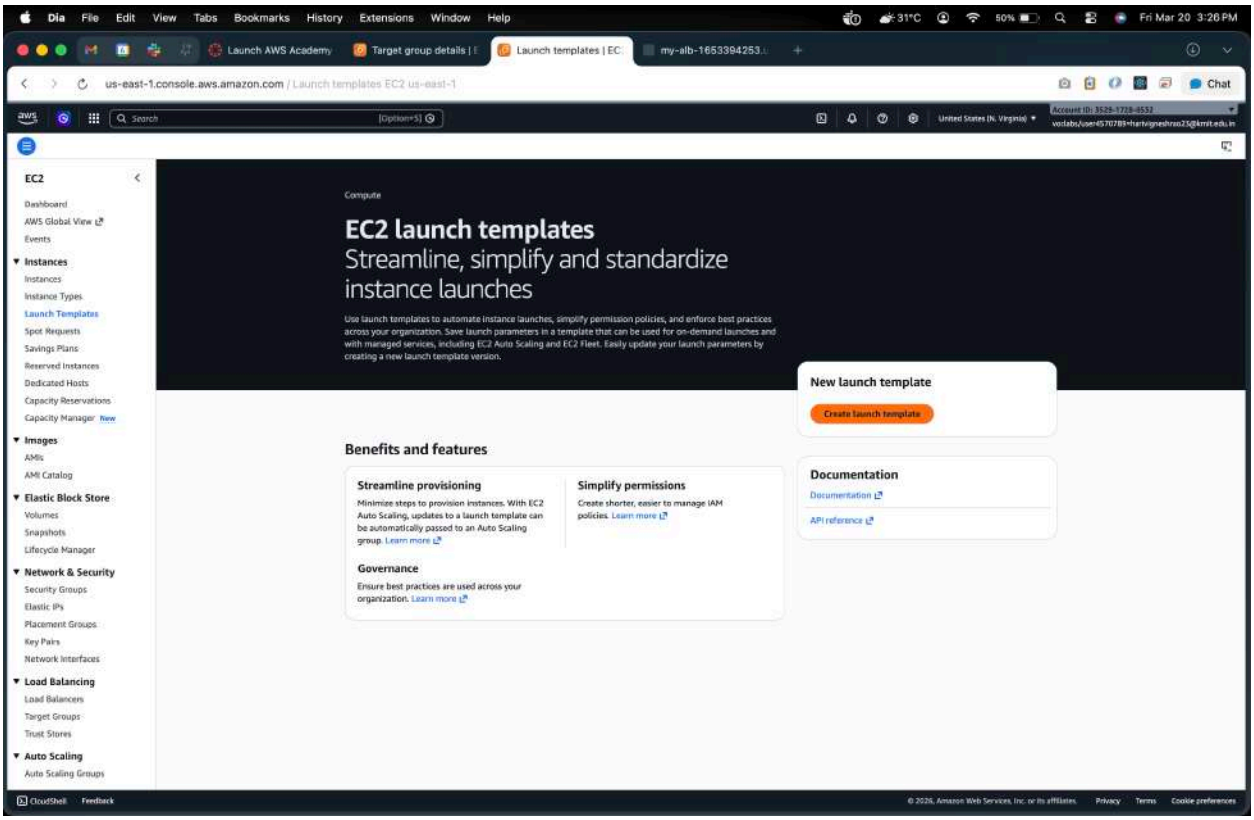
4. **Provide Image Name**: e.g., my-webserver-AMI
5. Optional: Add description
6. Click **Create Image**
7. Wait for the AMI to become **available** (status: available in **AMIs** section)

This AMI will be used for launching new instances automatically in ASG.

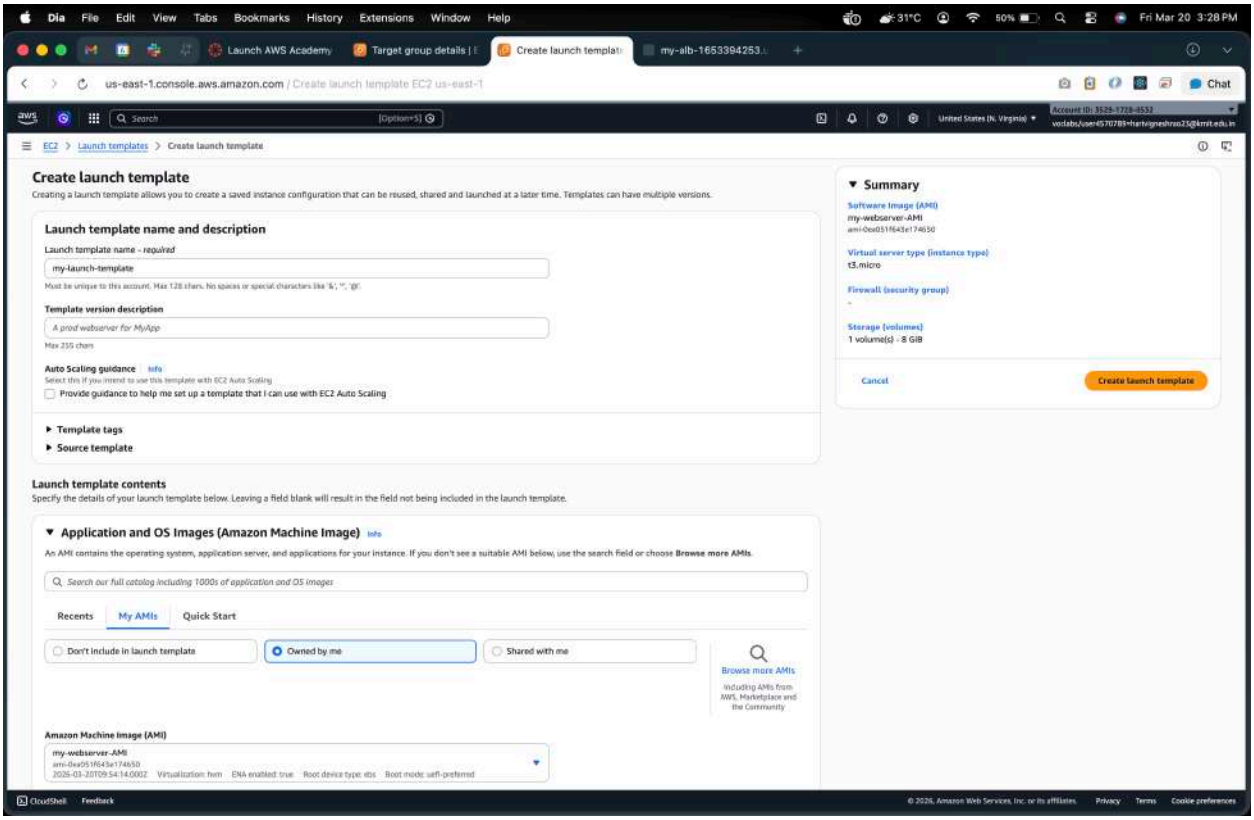


Step 2: Create a Launch Template

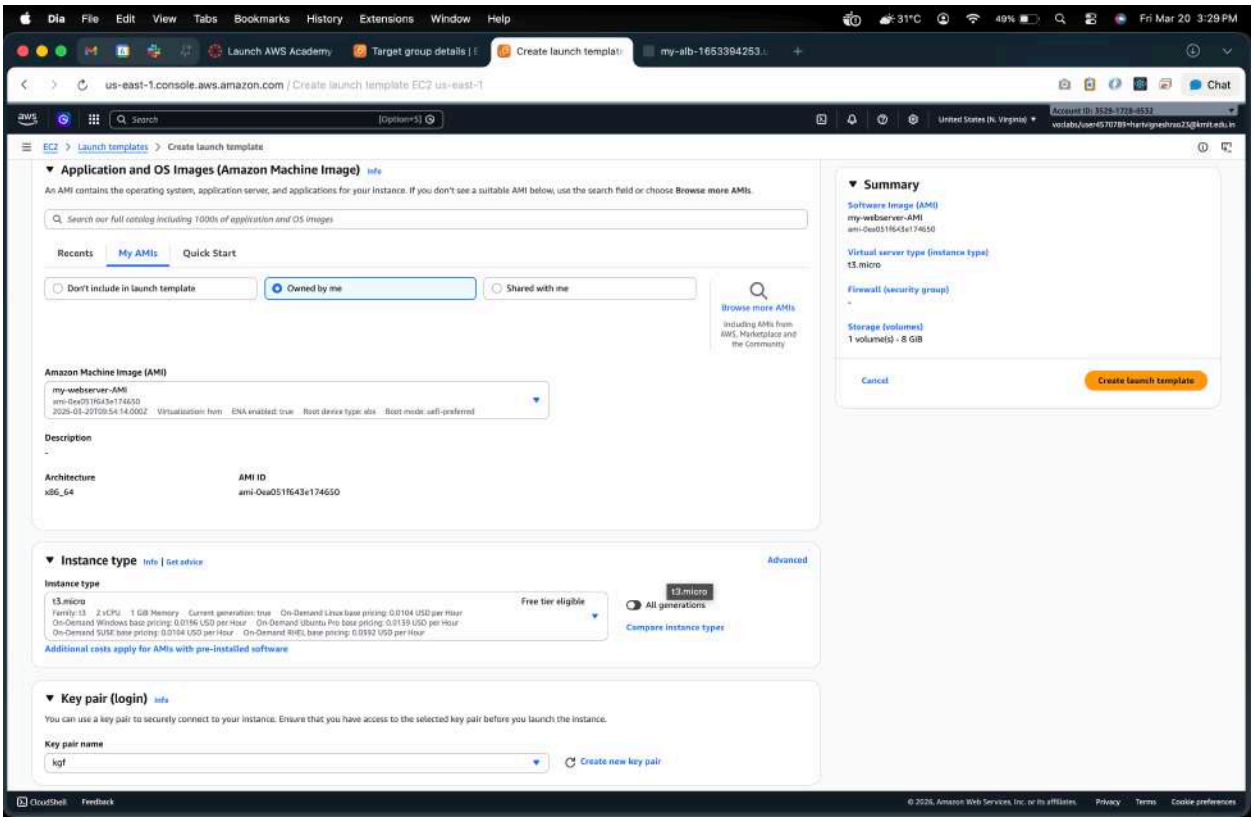
1. Go to **EC2 → instances->Launch Templates → Create Launch Template**



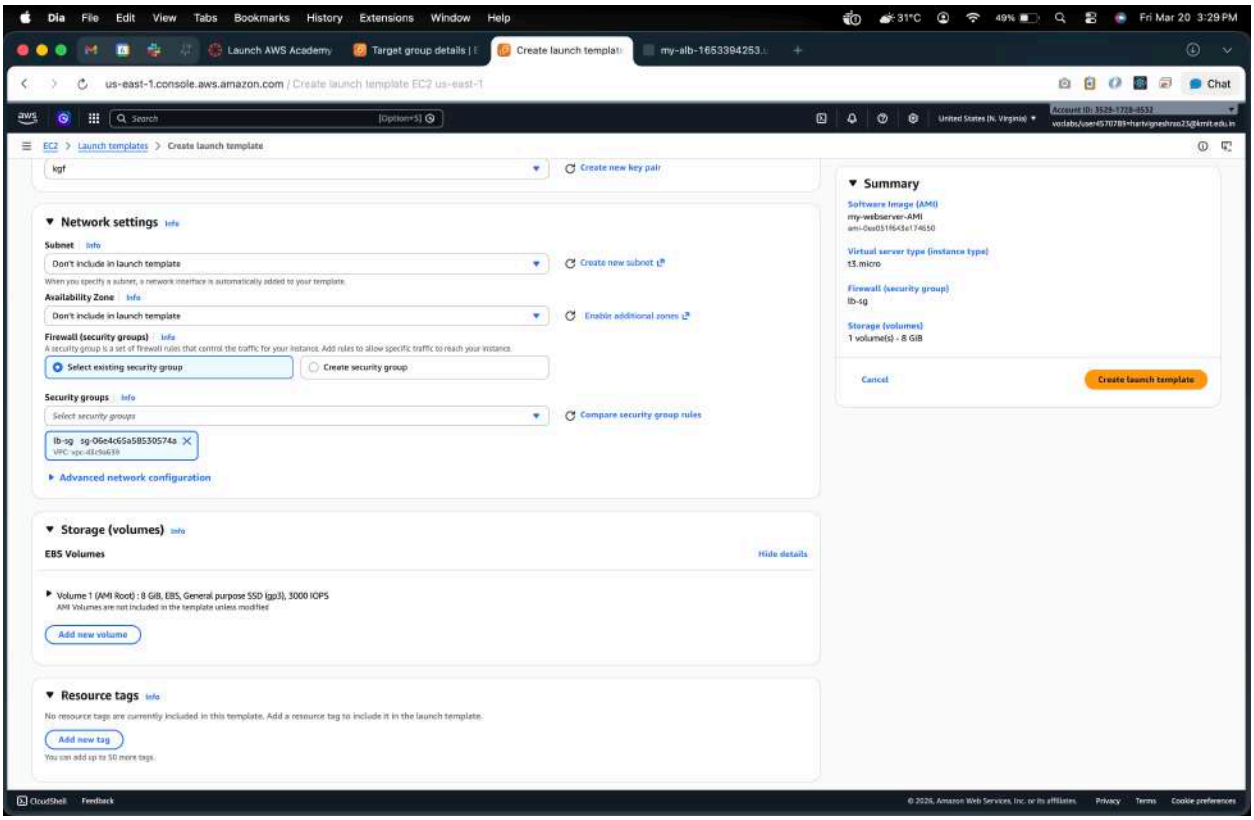
2. **Launch Template Name**: my-launch-template



3. **AMI**: Select the AMI created in Step 1 (my-webserver-AMI)
4. **Instance Type**: t2.micro (or your choice)
5. **Key Pair**: Select your existing key pair

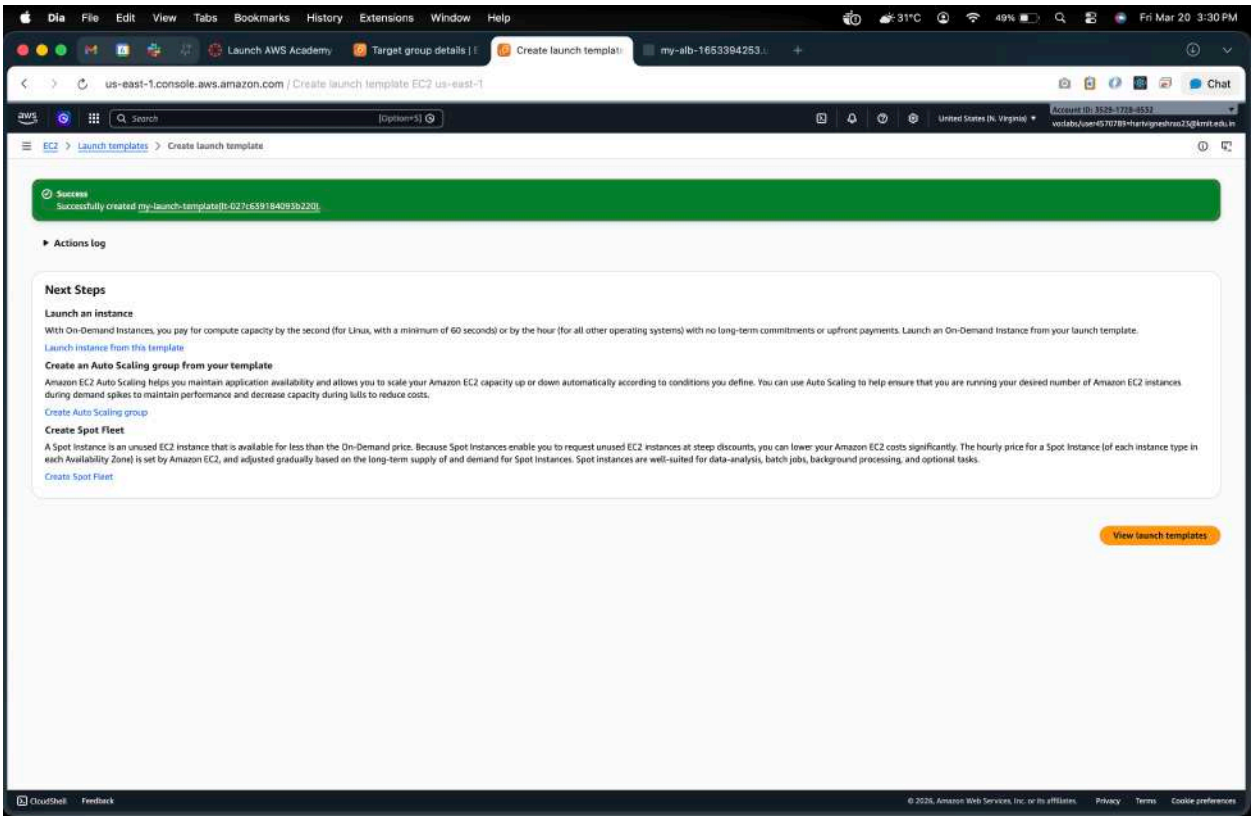


6. **Security Groups**: Select the **Load Balancer Security Group** (allow HTTP 80)



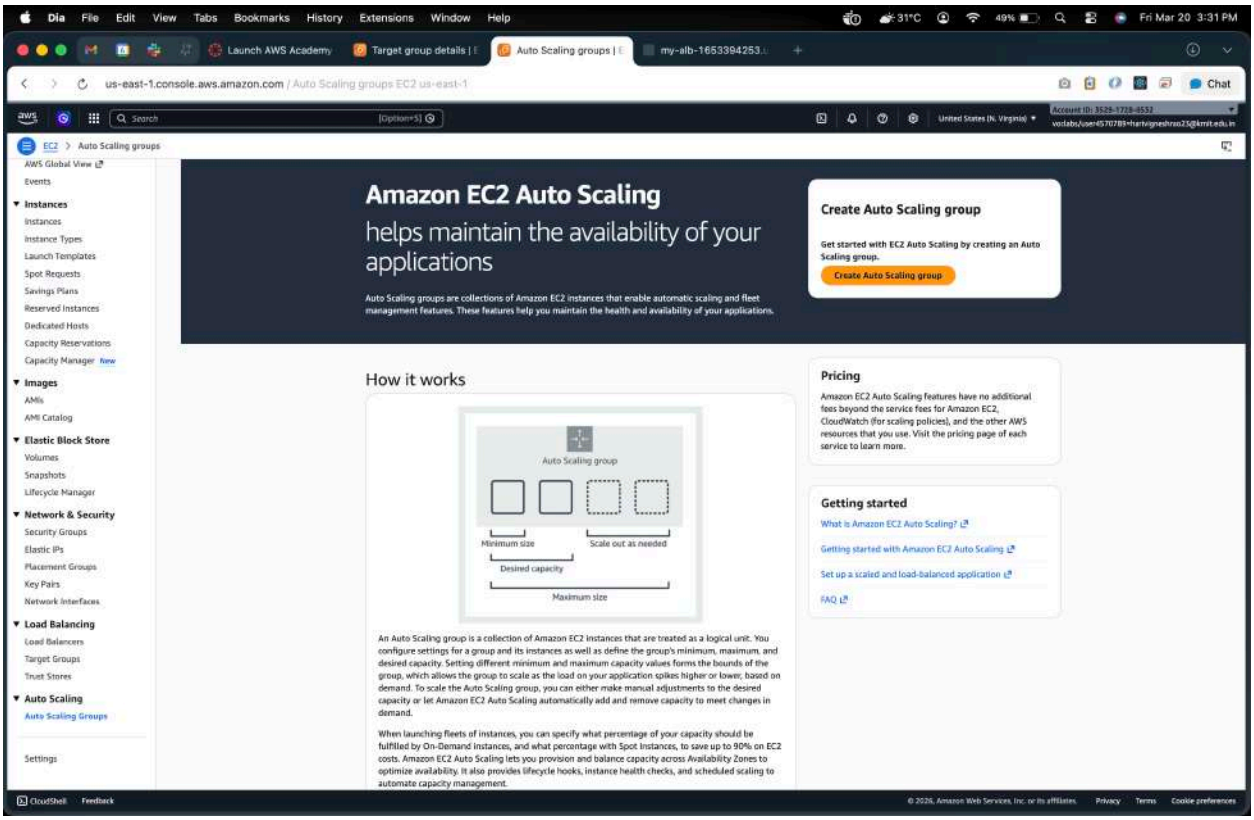
7. **User Data**: Optional (already baked into AMI)
8. Click **Create Launch Template**

Launch Template is ready for Auto Scaling.

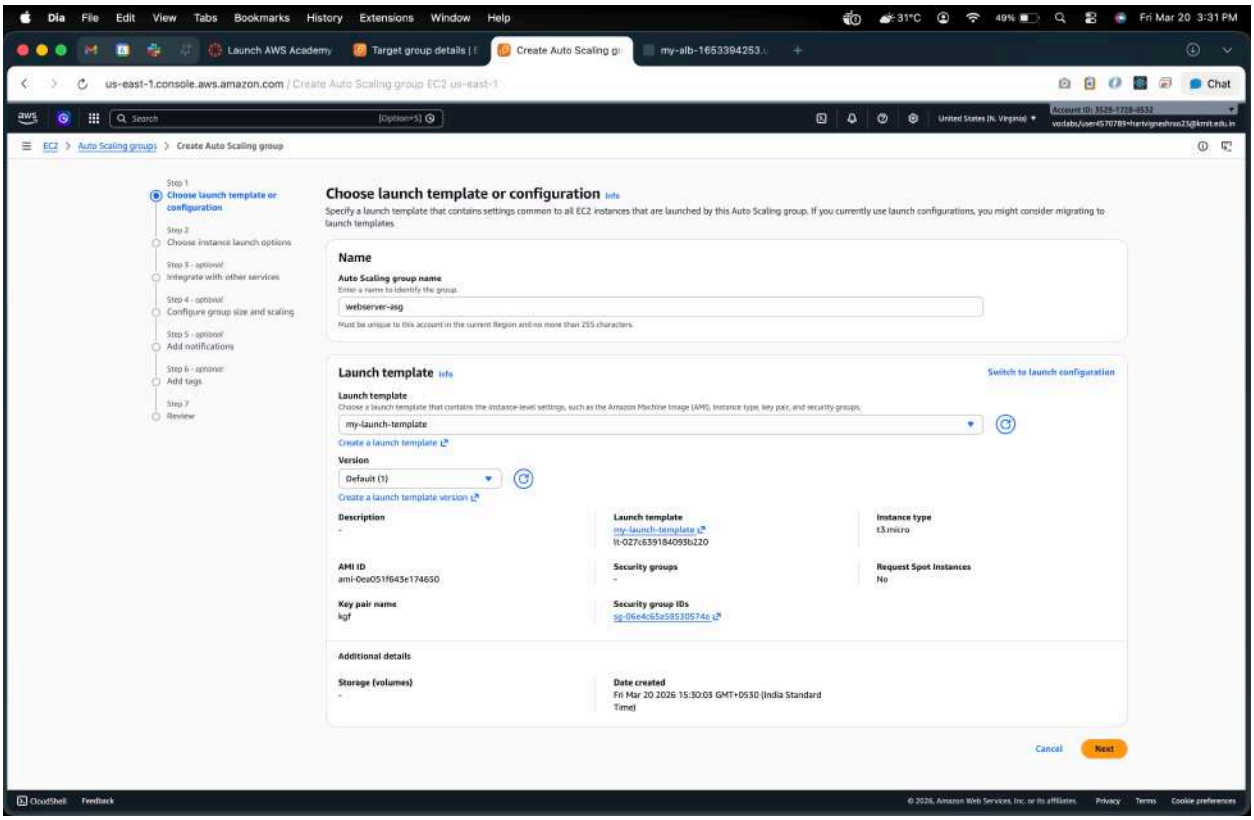


Step 3: Create Auto Scaling Group (ASG)

1. Go to **EC2 → Auto Scaling Groups → Create Auto Scaling Group**

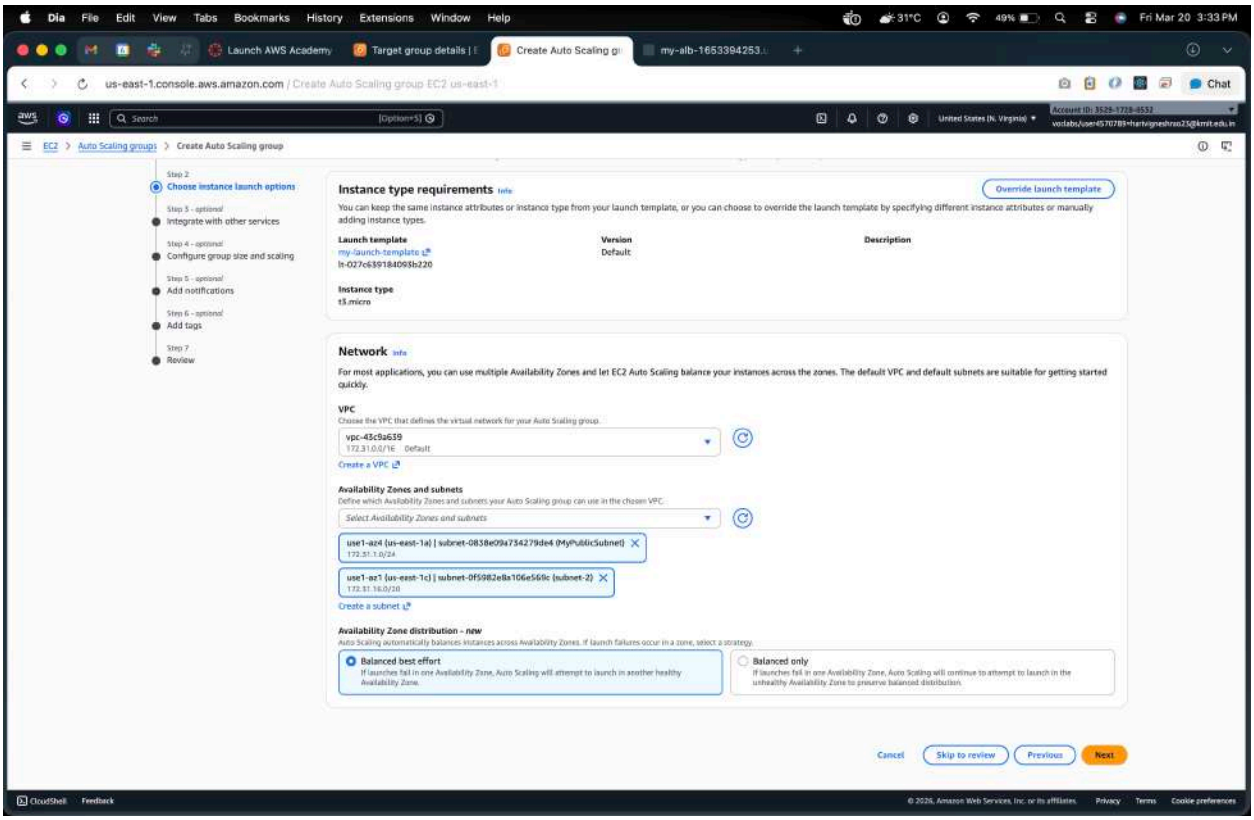


2. **Name**: webserver-asg
3. **Launch Template**: Select my-launch-template
4. **Template Version**: Select **Version 1**
5. Click **Next**

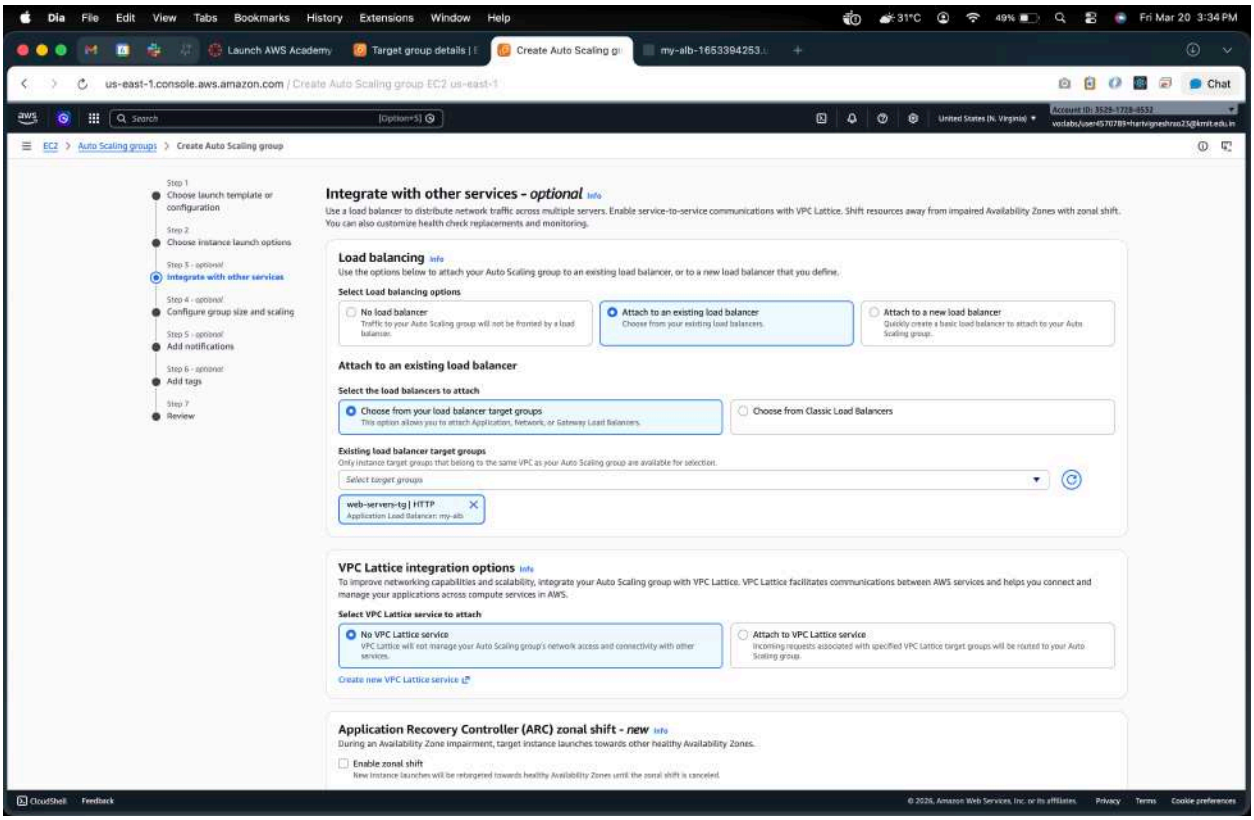


Step 3.1: Network Settings

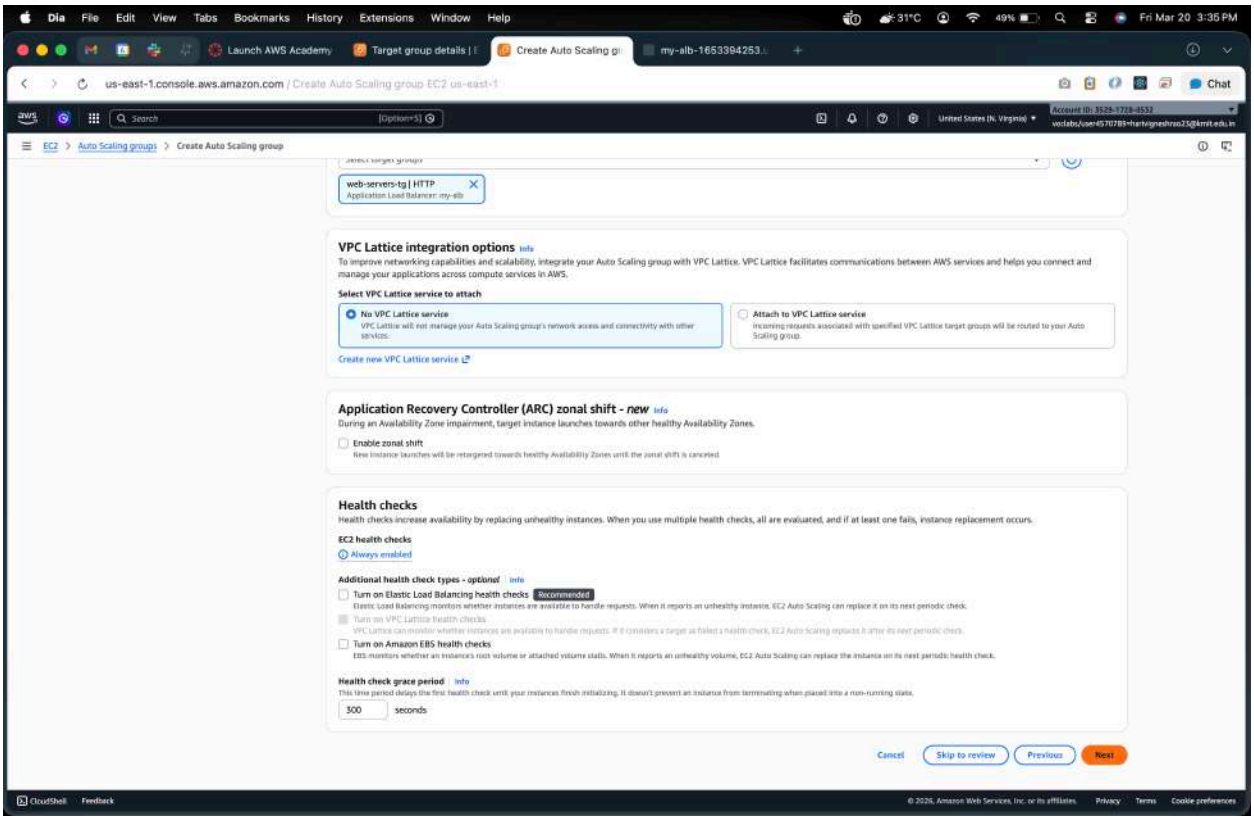
1. **VPC**: Default VPC (or your existing VPC)
2. **Availability Zones (AZs)**: Select **all** subnets (for high availability)



3. **Load Balancer**: Enable **Attach to an existing load balancer**
    - Select your **existing Load Balancer**

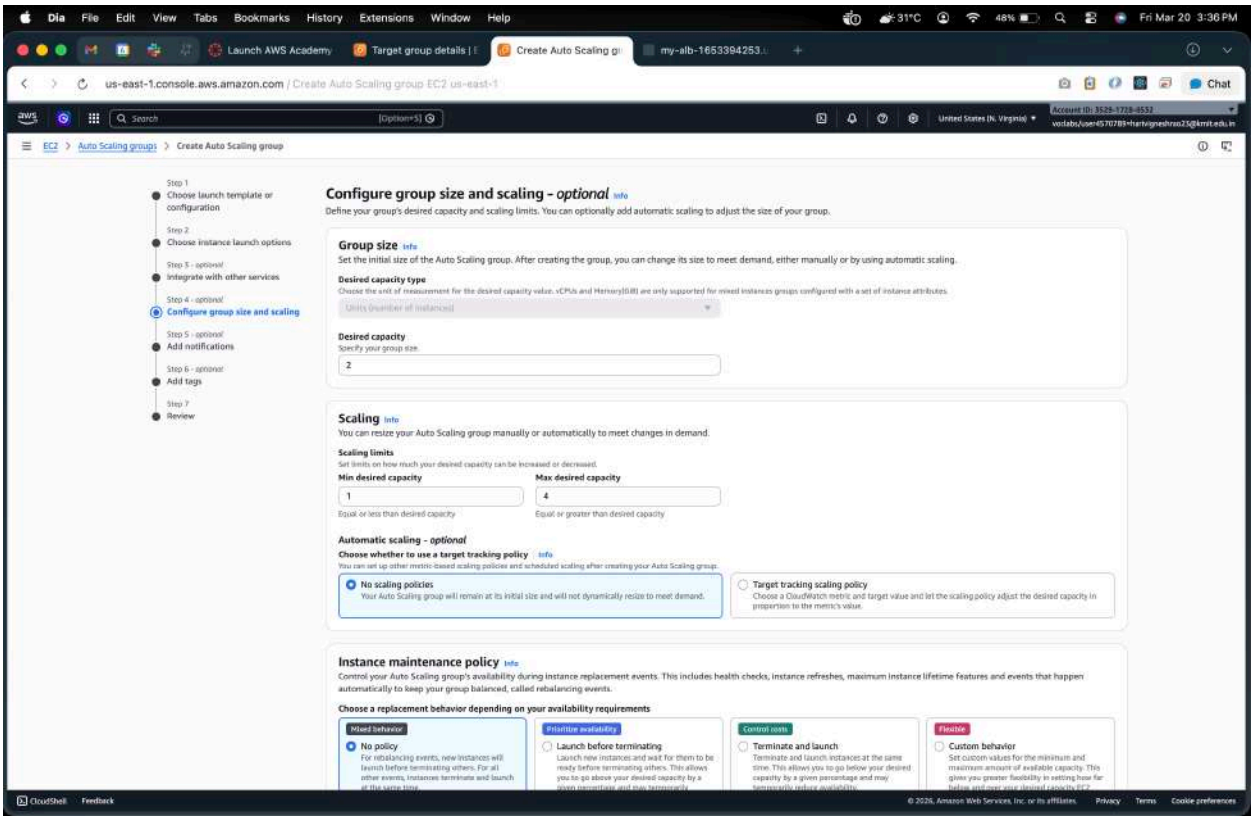


4. **Health Check**: HTTP
    - Set **Health Check Grace Period**: 300 seconds
5. Click **Next**

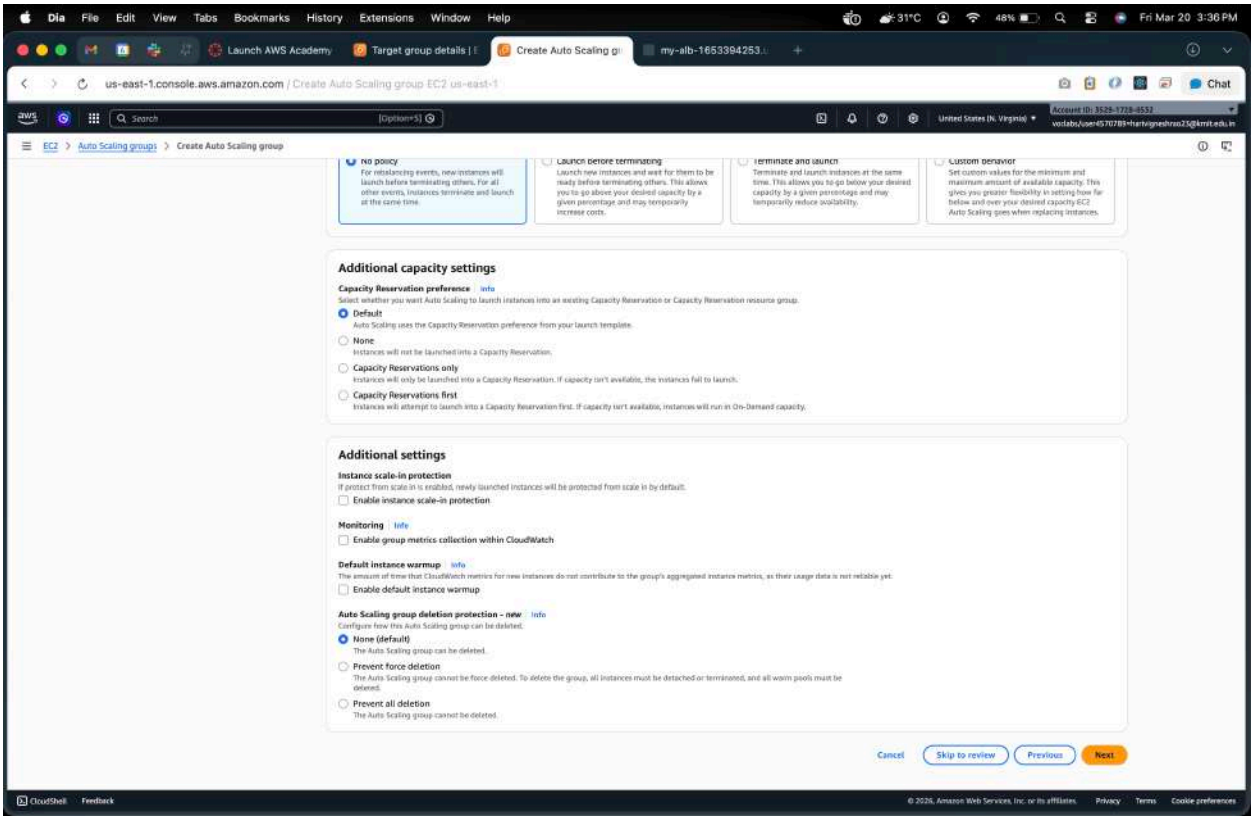


Step 3.2: Configure Group Size

1. **Desired Capacity**: 2 (initial instances)
2. **Minimum Capacity**: 1
3. **Maximum Capacity**: 4


Configure group size and scaling - optional
Group size
Desired capacity
2
Scaling
Min desired capacity
1
Max desired capacity
4
No scaling policies
Target tracking scaling policy
Instance maintenance policy
No policy
Launch before terminating
Terminate and launch
Custom behavior

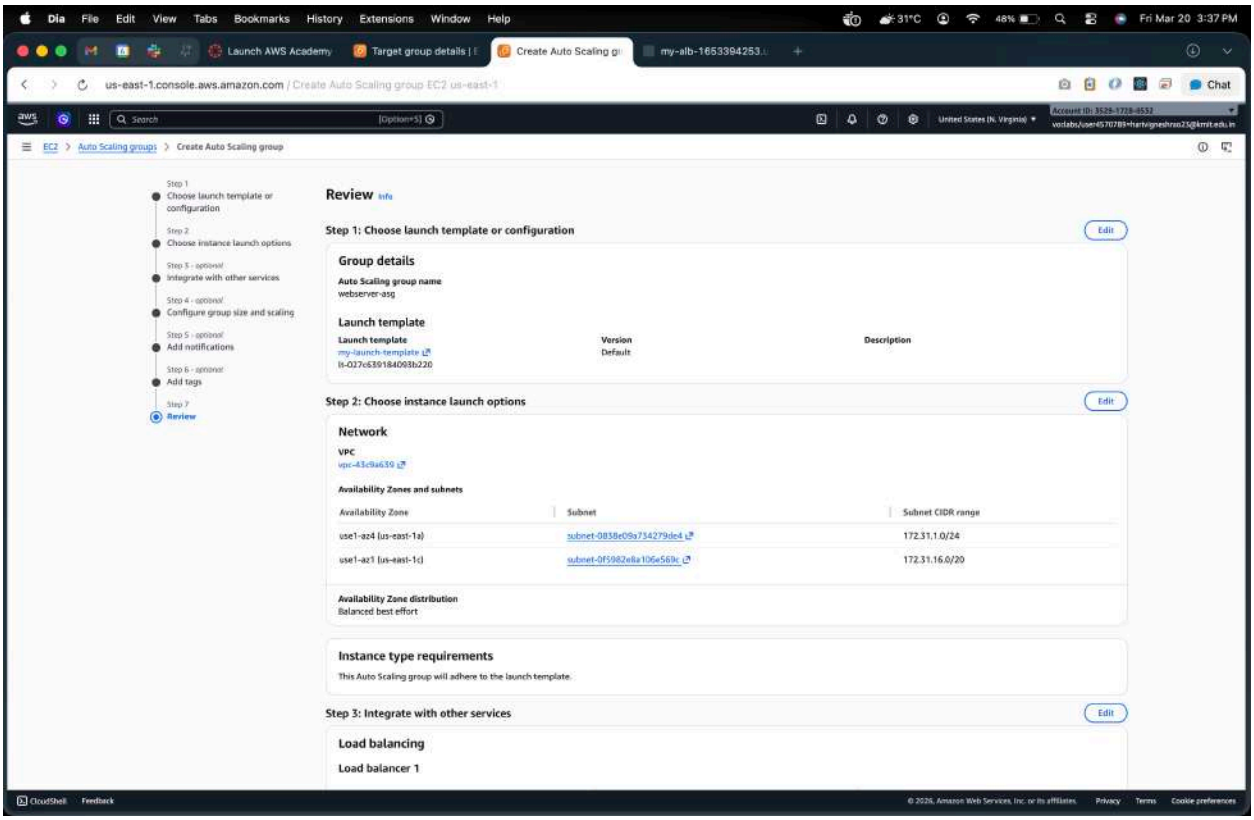
4. Click **Next**

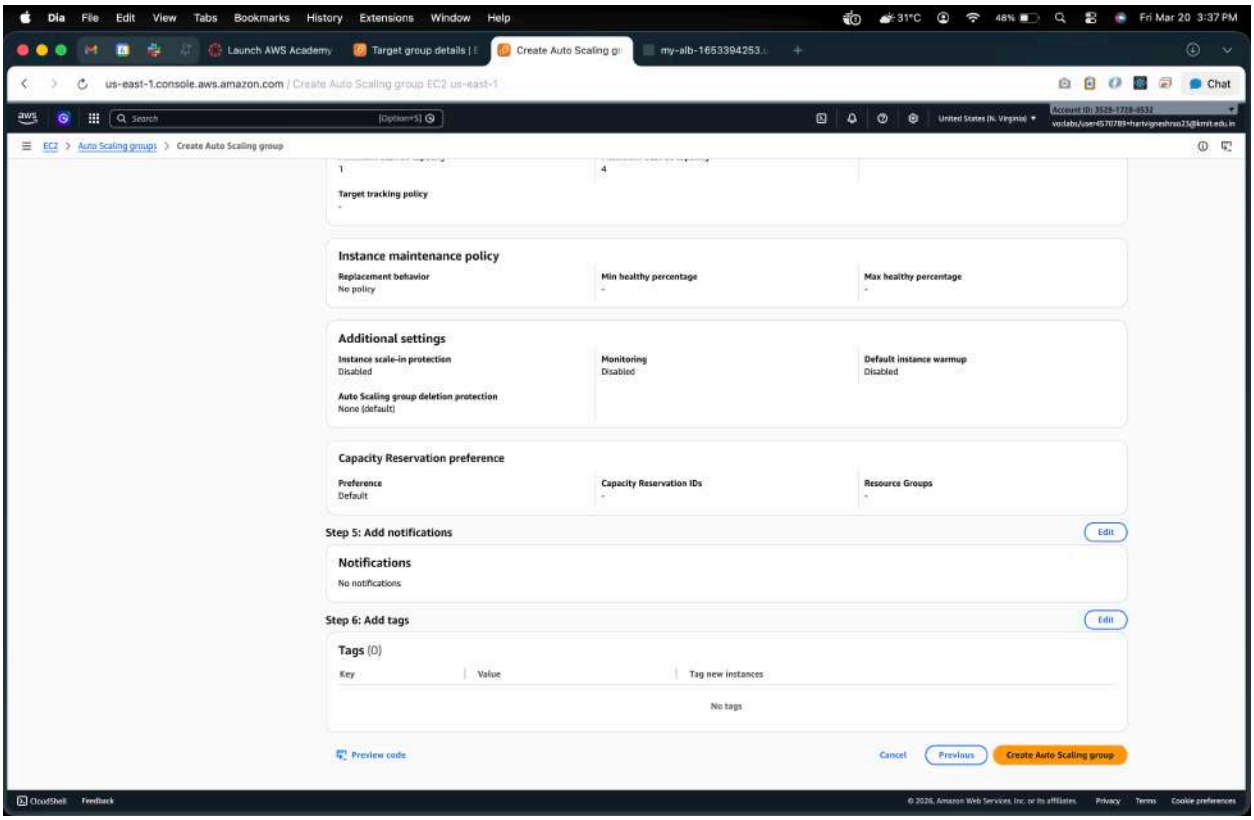

Additional capacity settings
Default
None
Capacity Reservations only
Capacity Reservations first
Additional settings
Enable instance scale-in protection
Enable group metrics collection within CloudWatch
Enable default instance warmup
None (default)
Prevent force deletion
Prevent all deletion
Cancel
Skip to review
Previous
Next

Step 3.3: Review and Create ASG

1. Review all settings
2. Click **Create Auto Scaling Group**

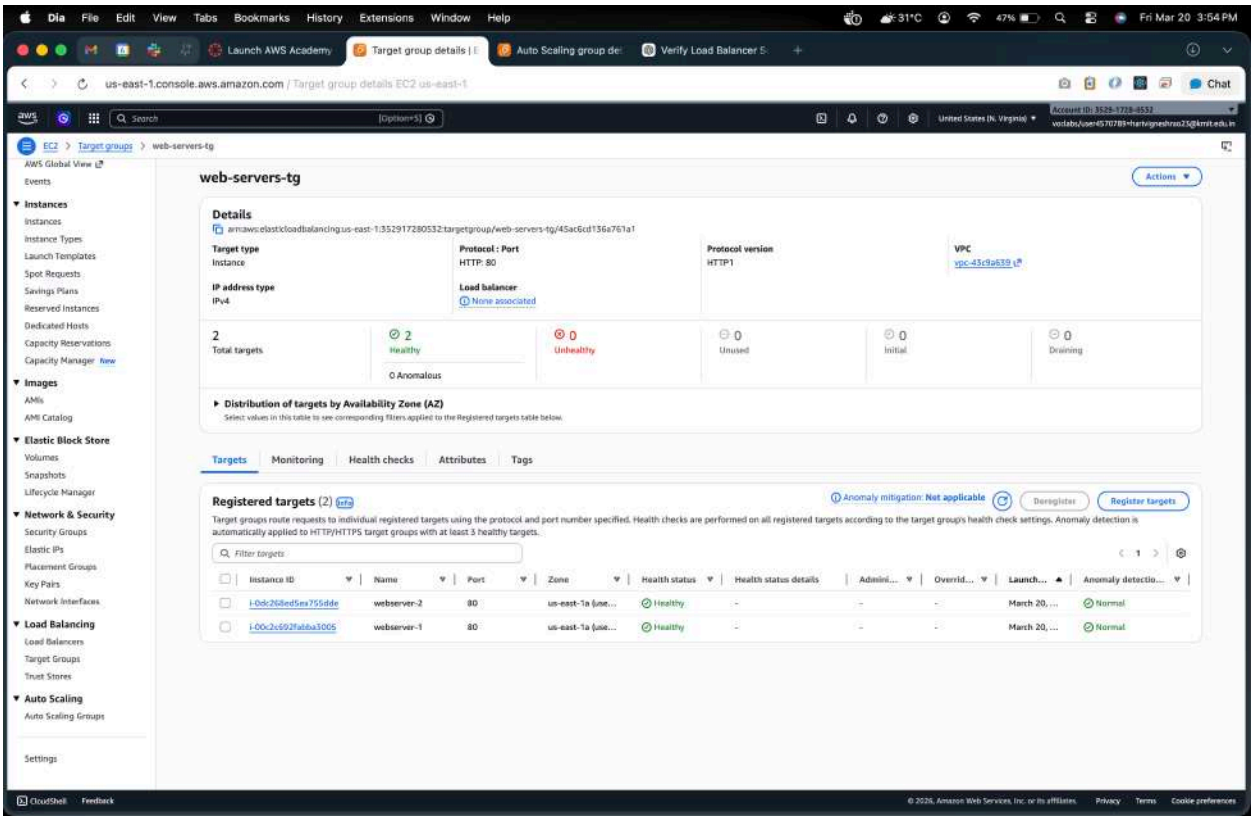
Auto Scaling Group is now ready and attached to your Load Balancer.



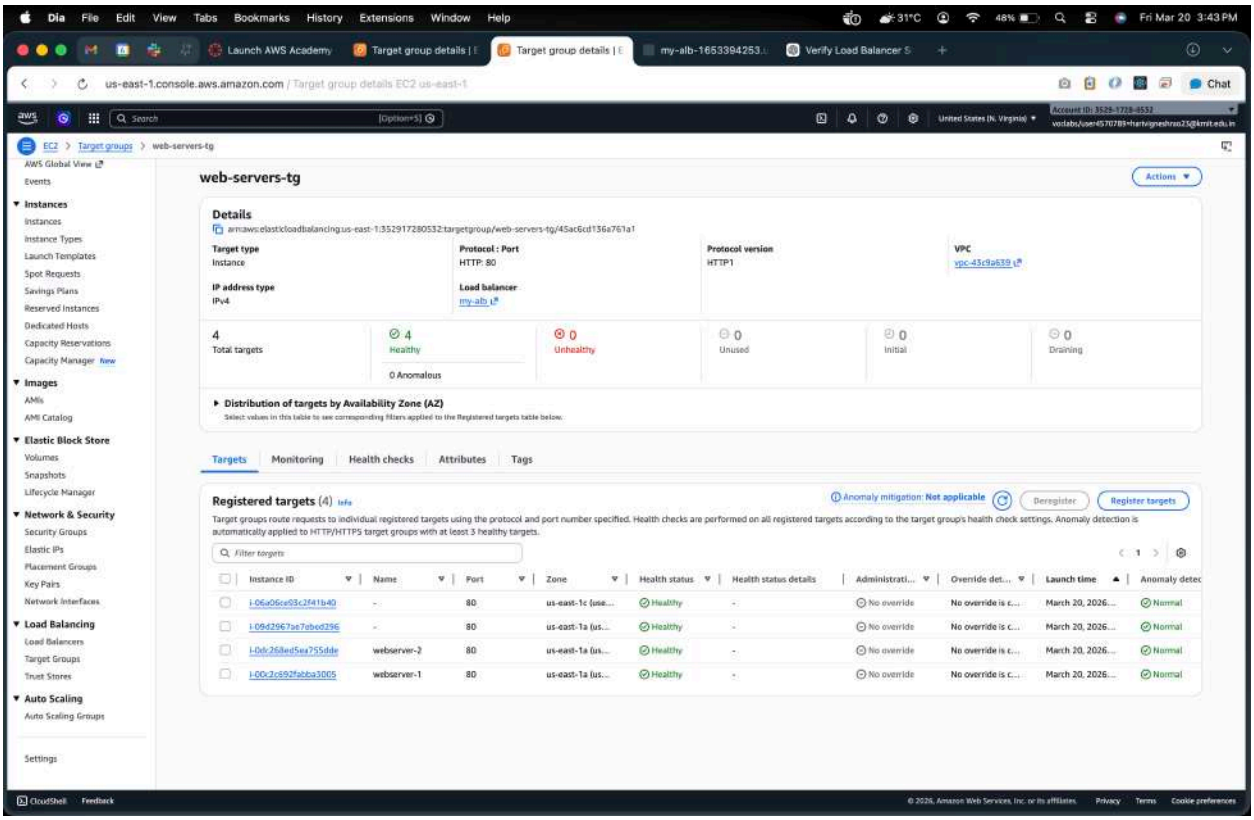


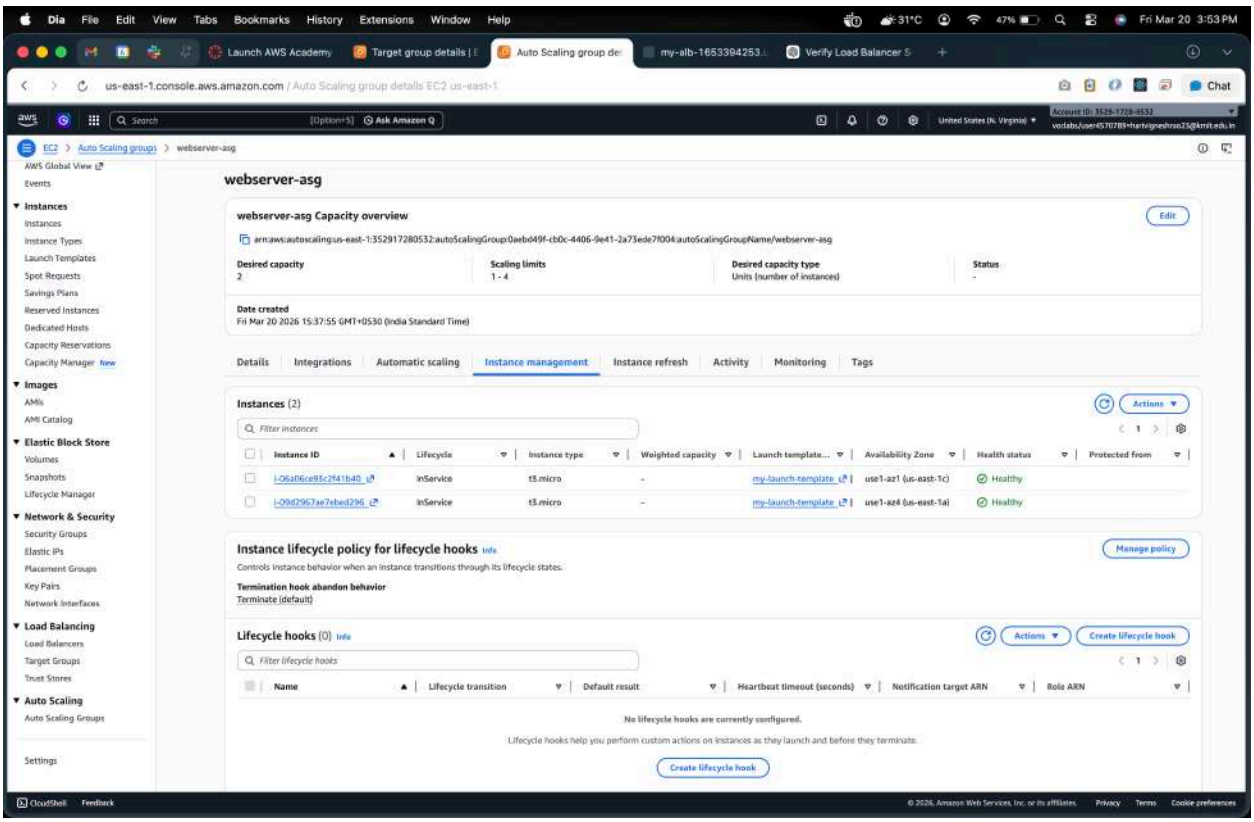
Step 4: Verify Load Balancer and Auto Scaling

1. Go to **Load Balancers → Select your Load Balancer**
2. Click **Instances → Target Group → Registered Targets**
    - You will see **2 healthy instances** initially



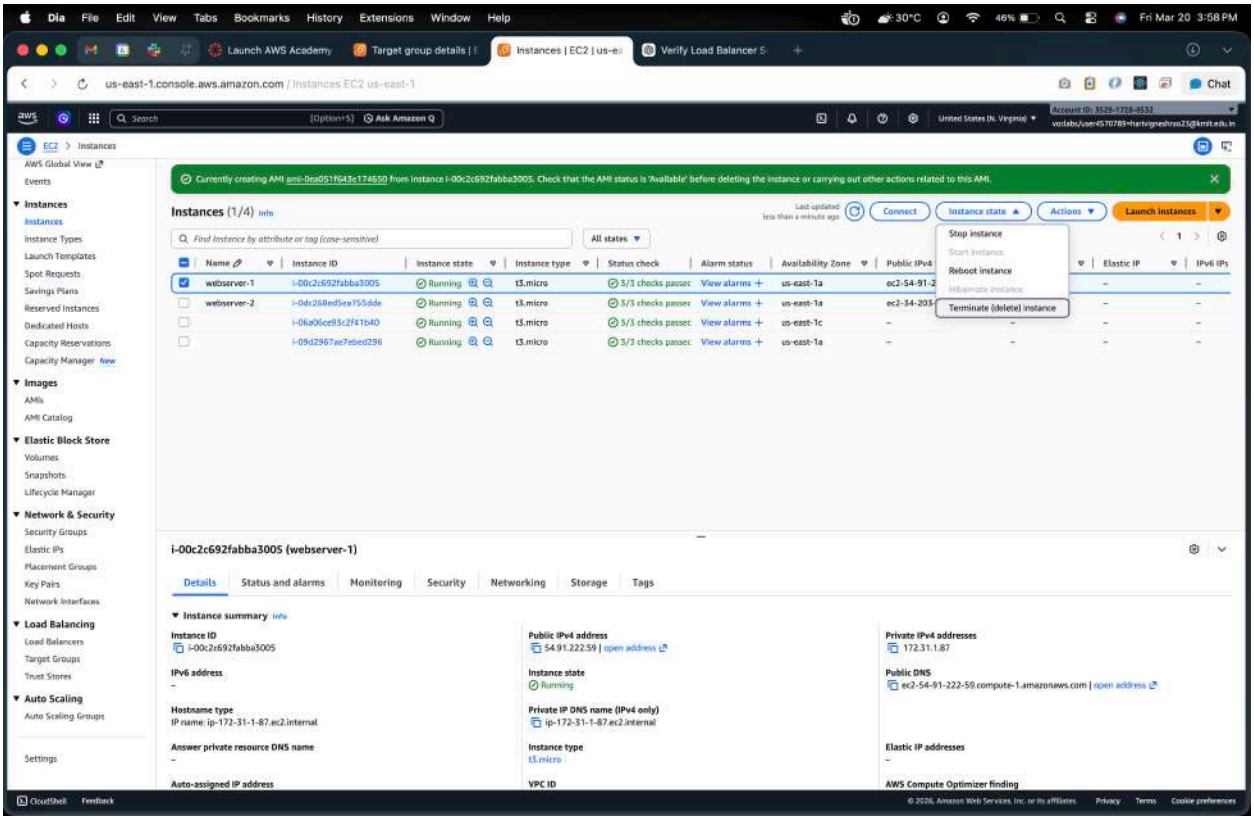
3. Check **Auto Scaling Group → Instances**
    - You may see up to **4 instances** if scaling triggers



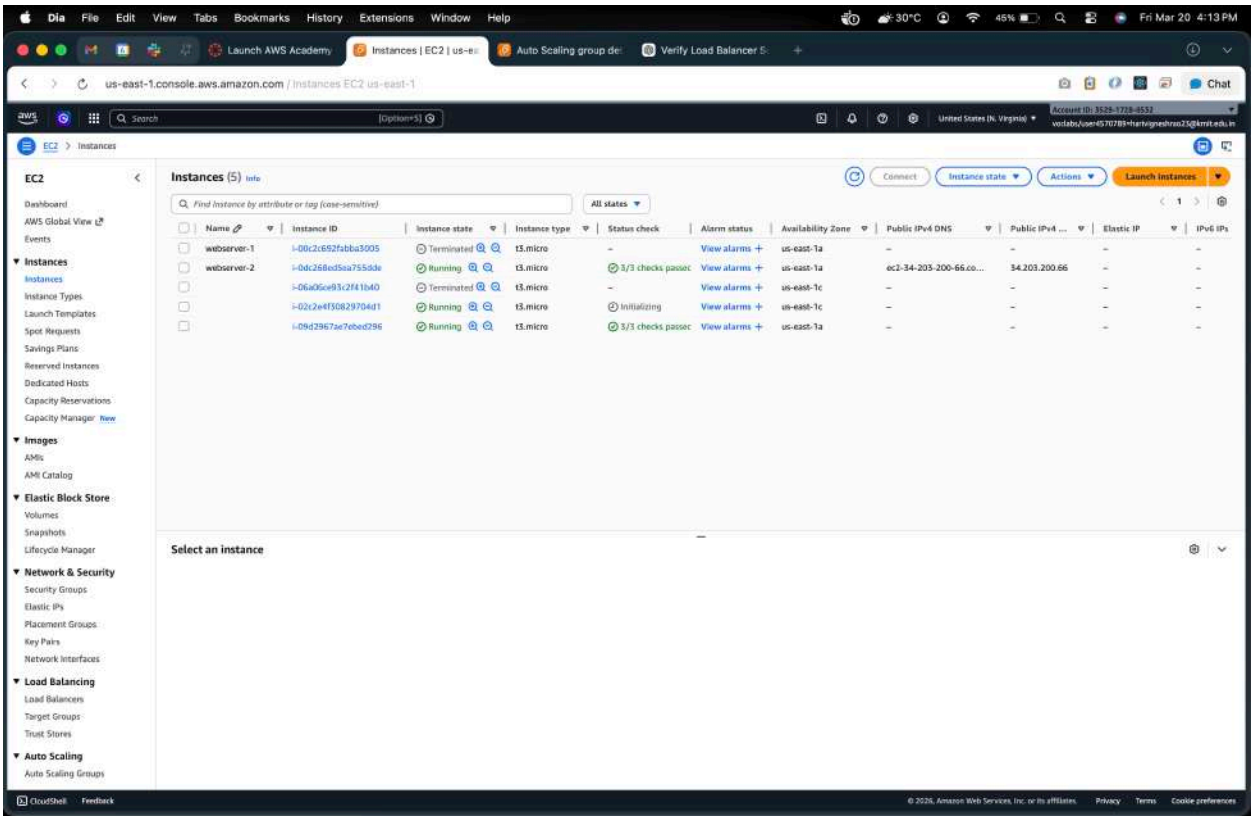


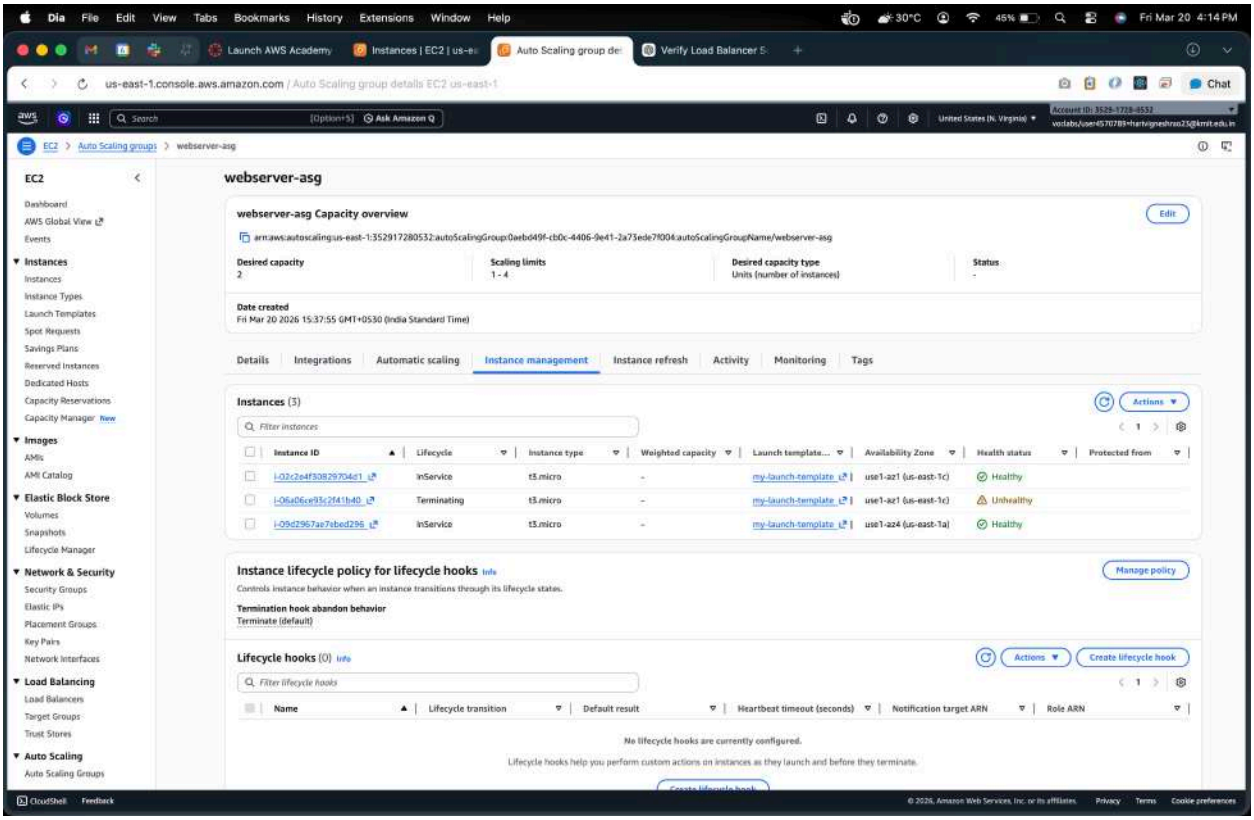
4. Test **auto recovery:**

- **Terminate one of the instances**



- After a few seconds, **ASG will launch a new instance automatically**
- Traffic will automatically be **distributed by the Load Balancer**

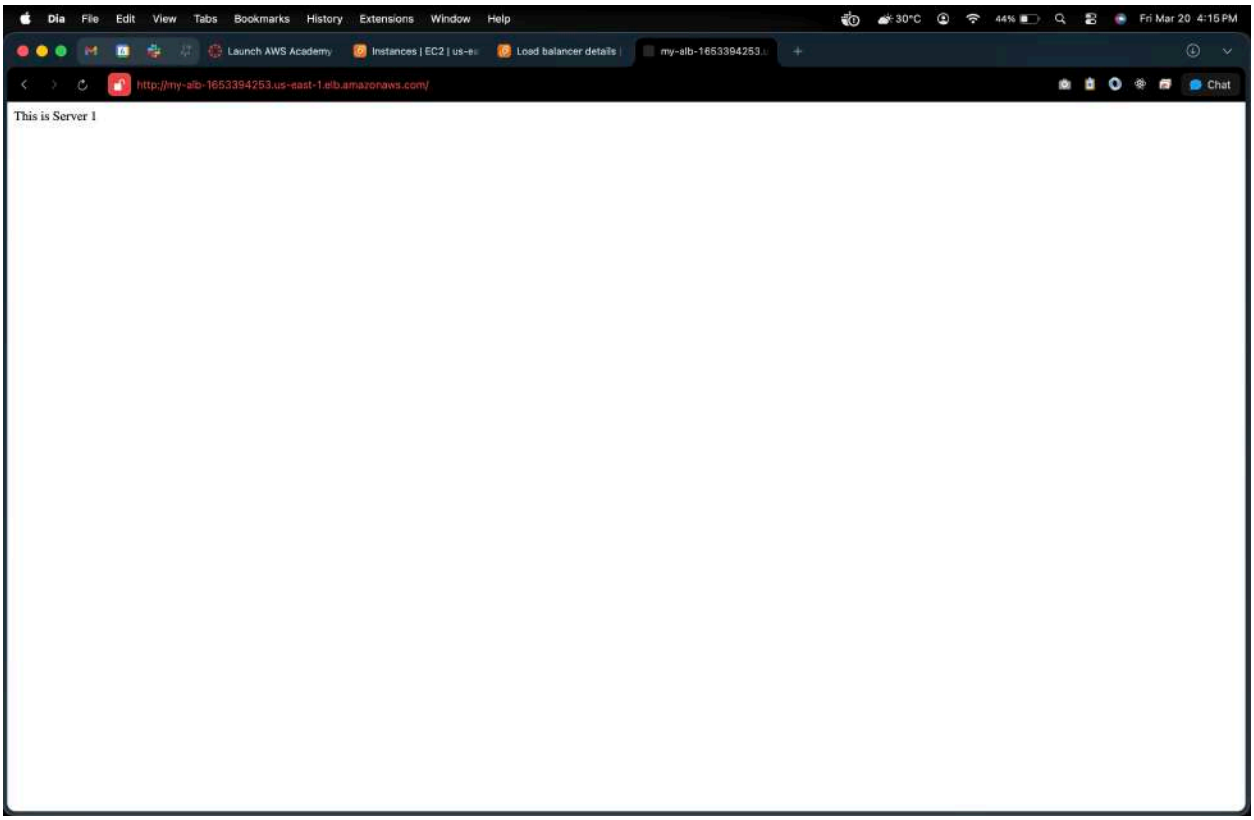


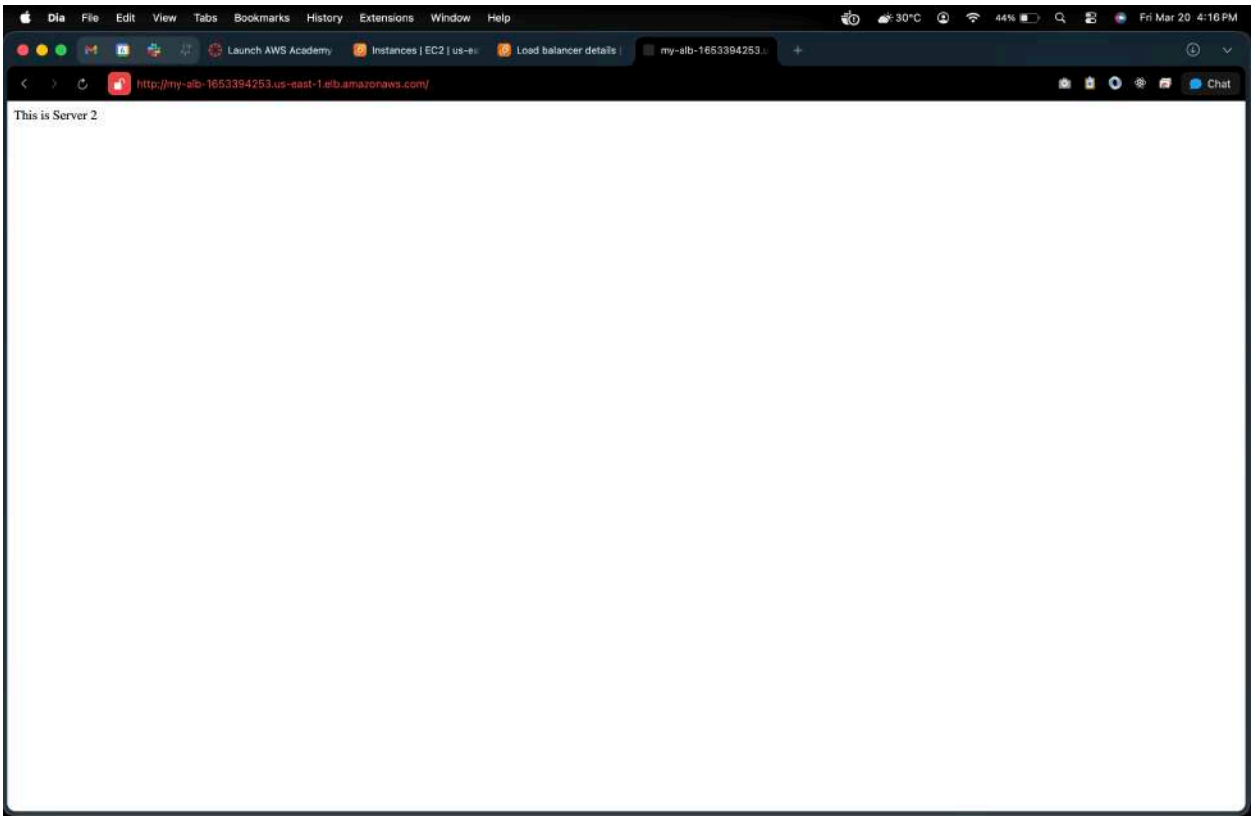


Step 5: Test the Load Balancer

1. Copy the **Load Balancer DNS Name** from ALB
2. Open in a browser → You should see your web page from one of the instances
3. Refresh multiple times → traffic alternates between instances

This confirms **Auto Scaling + Load Balancer** is working correctly.





Summary of Flow

1. **Create AMI** → snapshot of working EC2
2. **Create Launch Template** → using AMI, security group
3. **Create Auto Scaling Group** → attach Launch Template, set min/desired/max, attach Load Balancer
4. **Verify instances** → ASG launches/replaces instances automatically
5. **Test Load Balancer** → distributes traffic between all active instances